

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

**NVGo: Mobile Application for Efficient Communication and
Service**

Management in Nueva Valencia

**In partial fulfillment for the requirements in Capstone 1 and
Research Subject**

Sayo, Clark Eldine- BSIT 3-6

Pami, John Mark-BSIT3-6

Tamayo, Oyo Frederick - BSIT3-6

Viñarta, Jude Ryan - BSIT3-7

Rodriguez, Adrian - BSIT3-7

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Table of Contents

CHAPTER 1: THE PROBLEM AND ITS BACKGROUND Page 1-30

1.1 Introduction	1
1.2 Background of the Study	3
1.3 Statement of the Problem	6
1.4 Objectives of the Study	8
1.5 Significance of the Study	10
1.6 Scope and Limitation of the Study	13
1.7 Definition of Terms	15
1.8 Conceptual Framework	18
1.9 Research Locale	21
1.10 Organization of the Study	23
1.11 Assumptions of the Study	25
1.12 NVgo System Overview	27
1.13 Chapter Summary	29

*CHAPTER 2: REVIEW OF RELATED LITERATURE AND STUDIES . Page 31-110

2.1 Legal Bases and Government Policies	31
2.1.1 RA 7160 Local Government Code	31
2.1.2 RA 9485 Anti-Red Tape Act	32
2.1.3 RA 8792 E-Commerce Act	33
2.1.4 RA 10173 Data Privacy Act	34

2.1.5 EO 2 Freedom of Information	35
2.2 Communication and Information Dissemination in LGUs	36
2.2.1 Inefficiency of Traditional Channels	36
2.2.2 Problems with Social Media Groups	37
2.2.3 Lack of Two-Way Communication	38
2.2.4 Digital Divide in Rural Areas	39
2.3 Disaster Risk Reduction and Emergency Communication	40
2.3.1 Delayed Alerts in Island Municipalities	40
2.3.2 Best Practices in Emergency Notification	41
2.3.3 Case Study: Japan J-Alert System	42
2.3.4 NDRRMC Guidelines	43
2.4 Service Delivery and Citizen Feedback Systems	44
2.4.1 Manual vs Digital Complaint Systems	44
2.4.2 Ticketing Systems in Public Service	45
2.4.3 Impact of Tracking on Citizen Satisfaction	46
2.4.4 SLA and Response Time in LGUs	47
2.5 Mobile Applications for Government	48
2.5.1 Mobile First Strategy	48
2.5.2 UI/UX Design for Rural Users	49
2.5.3 Language Localization	50
2.5.4 Retention and Adoption Factors	51
2.6 Existing LGU Mobile Applications in the Philippines	52
2.6.1 eGovPH - National Level	52
2.6.2 Iloilo City 311 - City Level	53
2.6.3 eBarangay - Barangay Level	54

2.6.4 Passi City App, Antique App, Kabankalan App	55
2.7 International Practices and Case Studies	56
2.7.1 Boston 311 System	56
2.7.2 NYC 311 System	57
2.7.3 Singapore http://Gov.sg App	58
2.7.4 Smart Governance in ASEAN	59
2.8 Theories and Models	60
2.8.1 Technology Acceptance Model - TAM	60
2.8.2 Unified Theory of Acceptance - UTAUT	61
2.8.3 DeLone & McLean IS Success Model	62
2.8.4 Communication Theory	63
2.9 Security, Privacy, and Technical Standards	64
2.9.1 Data Privacy Compliance	64
2.9.2 Cybersecurity for LGU Apps	65
2.9.3 Firebase and Cloud Services	66
2.9.4 Authentication and Data Protection	67
2.10 Development Methodologies	68
2.10.1 Agile Methodology	68
2.10.2 SDLC for Government Systems	69
2.10.3 Usability Testing Standards	70
2.10.4 System Quality Model ISO 25010	71
2.11 Impact of Digital Governance	72
2.11.1 Transparency and Accountability	72
2.11.2 Citizen Participation	73
2.11.3 LGU Performance Metrics	74

2.11.4 Seal of Good Governance	75
2.12 Synthesis and Gap Analysis	76
2.12.1 Summary of Related Literature	76
2.12.2 Comparison Table of Existing Systems	78
2.12.3 Research Gap	80
2.12.4 How NVgo Fills the Gap	82
2.13 Theoretical Framework	84
2.13.1 UTAUT Model Diagram and Explanation	84
2.13.2 DeLone & McLean Model Diagram and Explanation	86
2.14 Conceptual Framework	88
2.14.1 IPO Model Diagram	88
2.14.2 Input-Process-Output Explanation	89
2.15 Local Studies Review	90
2.15.1 Studies in Region 6	90
2.15.2 Studies in Guimaras	92
2.15.3 Studies in Iloilo	94
2.16 Foreign Studies Review	96
2.16.1 Studies in Asia	96
2.16.2 Studies in Europe and US	98
2.16.3 International Standards	100
2.17 Chapter Summary	102
[Continue RRL Sources 109-138 with 1 page each = 103-110]	
*CHAPTER 3: METHODOLOGY	Page 111-150
3.1 Research Design	111
3.2 System Development Methodology	113

3.3 System Architecture	115
3.3.1 Three-Tier Architecture Diagram	115
3.3.2 Technology Stack	116
3.3.3 Explanation	117
3.4 Data Flow Diagram	118
3.4.1 Context Diagram Level 0	118
3.4.2 DFD Level 1	119
3.4.3 DFD Level 2 - Service Request	120
3.4.4 Data Dictionary	121
3.4.5 Explanation	122
3.5 Use Case Diagram	123
3.5.1 Use Case Diagram	123
3.5.2 Use Case Descriptions Table	124
3.5.3 Explanation	125
3.6 Entity Relationship Diagram	126
3.6.1 ERD Diagram	126
3.6.2 Data Dictionary	127
3.6.3 Explanation	128
3.7 System Features and Functions	129
3.7.1 Announcement Module	129
3.7.2 Service Request Module	130
3.7.3 Emergency Alert Module	131
3.7.4 Admin Dashboard	132
3.8 System Requirements	133
3.8.1 Hardware Requirements	133

3.8.2 Software Requirements	134
3.8.3 Network Requirements	135
3.9 Testing and Evaluation	136
3.9.1 System Testing Procedure	136
3.9.2 Evaluation Instrument - SUS	137
3.9.3 Respondents of the Study	138
3.9.4 Statistical Treatment	139
3.10 Implementation Plan	140
3.11 Ethical Considerations	141
3.12 Project Timeline	142
3.13 Chapter Summary	143
REFERENCES	144
APPENDICES	148
Appendix A: Survey Questionnaire	148
Appendix B: Letters	149
Appendix C: Screenshots	150

CHAPTER 1

INTRODUCTION

1. 1 Background of the Study

The rapid growth of modern technology in the Philippines has greatly changed the way people communicated with each other, which made socialization convenient for people. However, despite this convenience, not every island in the Philippines could catch up to the modern technological advancements due to their environmental challenges and remote location.

Hon. Emmanuel C. Galila, the mayor of the municipality of Nueva Valencia, Guimaras, recently partnered with the Department of Information and Communications Technology (DICT). The agency deployed Starlink internet connections throughout Nueva Valencia, especially across its remote islands, enabling the local government to connect with residents online.

Despite the improved accessibility in communication, several limitations persisted. The municipality remained vulnerable to disinformation, and officials needed to ensure that transmitted and received information was secure and legitimate. The mere availability of internet access did not guarantee a safe exchange of information.

To resolve these challenges, the municipality requested the development of a mobile application capable of verifying reports, dispatching the appropriate equipment, and improving information dissemination. Additional features were also integrated to facilitate convenient transactions for residents who were unable to visit the municipal hall personally. Thus, the proponents proposed

NVGo, a mobile application designed for efficient communication and service

management in Nueva Valencia. With Starlink deployed in the area, the application enhanced information dissemination and precise service management.

1.2 Statement of the problem

Despite the availability of internet connection in remote areas, several issues remain unresolved. This study seeks to address the following problems:

1. Insecure exchange of communication from both parties.
2. Illegitimate report on infrastructure damage.
3. Misrouted emergency calls that lead to improper responses and delays.
4. Fake profiles can be a threat and can cause false alarms.

1.3 Objectives of the study

1.3.1 General Objectives

The general objective of this study was to design and develop a reliable and convenient mobile application that improved citizen communication, facilitated proper emergency calls, and ensured legitimate profiles.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Designed a user friendly interface for information dissemination
2. Implement proper input reports and emergency response mechanisms.
3. Test the application's usability among citizens in remote areas.
4. Evaluate the effectiveness of the mobile application in improving convenience.
5. Improve municipal services.

1.4 Scope and Delimitation

The study involves the development of a mobile application utilizing React

(JavaScript) for the front-end and Express.js (Node.js) for the back-end. The core CRUD (Create, Read, Update, Delete) functionalities formed the initial phase of the development process. Prior to finalizing the mobile deployment, the application integrated real-time features such as notifications and live updates

The system featured user registration, profile creation, an admin dashboard, and real-time updates. The study did not entail large-scale data processing or sophisticated data analysis. The system was developed as a prototype to

demonstrate the feasibility and functionality of the proposed solution, focusing primarily on essential features. It was designed so that it could be further extended for real-world deployment.

1.5 Significance of the study

The proposed system will benefit the following:

- **Residents of Nueva Valencia** – It provided a streamlined platform for convenient complaint submission, incident reporting, and direct access to municipal announcements.
- **Municipality Officials** – It enabled efficient management, tracking, and monitoring of incoming public reports and local services.
- **Municipality of Nueva Valencia** – It enhanced overall service

delivery and improved operational efficiency within local governance.

- **Future Researchers** – It served as a valuable reference and foundation for subsequent studies on similar mobile-based municipal systems.
- **Proponents** – It offered practical experience in full-stack software

development, thereby enhancing technical and system design skills.

1.6 Definition of Terms

- **DICT** - DICT stands for the Department of Information and Communications Technology, the Philippine government agency responsible for the country's ICT development and digital transformation.
- **Starlink** - Starlink is a satellite internet service operated by SpaceX that provides high-speed, low-latency broadband globally, even in remote areas.
- **Information Dissemination** - is the process of distributing and sharing information to a targeted audience through various channels, ensuring clarity and accessibility.
- **Express.js**- is a JavaScript framework that helps build the server-side of a web application. It handles user requests, processes data, communicates with the database, and sends the appropriate responses to the frontend.
- **REACT** - is a JavaScript library created by Facebook for building user interfaces, particularly for single-page applications. It allows developers to create reusable UI components, which can manage their own state and be composed to build complex user interfaces.
- **Prototype**- an early, preliminary model or sample of a product built to test a concept, validate design ideas, and gather feedback before full- scale manufacturing or coding.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents relevant local and foreign literature and studies, alongside a synthesis of related literature, that provide the foundation framework for the study, **“NV Go: Mobile Application for Efficient Communication and Service Management in Nueva Valencia”**.

2.1 Foreign Studies

According to Janssen and Estevez (2016), mobile governance applications in Europe and Asia have significantly improved service delivery by reducing bureaucratic delays and enhancing transparency. Their study emphasized that mobile-first platforms are particularly effective in municipalities with limited resources, as they provide cost-efficient communication channels between citizens and government offices.

Similarly, Tapscott (2019) highlighted that mobile applications foster participatory governance by enabling real-time feedback, emergency alerts, and community engagement. In India, Sharma et al. (2020) found that mobile apps used for municipal service management increased citizen satisfaction and reduced response times for complaints and requests. These findings demonstrate the global relevance of mobile platforms in strengthening governance and service management.

Alqahtani & Kavakli (2016) – Proposed a mobile government framework in Saudi Arabia, showing that mobile apps enhance citizen engagement and service delivery efficiency (Journal of E-Government Studies).

Sharma, Singh, & Gupta (2018) – Studied emergency communication apps in India, integrating geolocation and push notifications to reduce response times (International Journal of Mobile Computing).

Gao & Bai (2020) – Examined mobile commerce apps in China, finding that service quality and trust strongly influence user retention (Journal of Retailing and Consumer Services).

Kumar & Raj (2022) – Investigated mobile health apps in Singapore, highlighting how real-time communication improves patient service management (Health Informatics Journal).

A Mobile App for Citizen Participation (2014)

Carolyn Schroeder (2014) explored the use of mobile applications to enhance citizen participation in local governance. The study found that mobile apps provide convenient and immediate channels for citizens to share opinions, participate in community discussions, and engage with local government initiatives. These findings are relevant to NVGo, which seeks to strengthen citizen involvement through digital communication.

The Acceptance of Mobile Government from a Citizens' Perspective (2014)

Jakob Ohme (2014) examined the factors influencing citizens' acceptance of mobile government services. The study found that mobile government applications have significant potential to improve communication between

governments and citizens. It emphasized that perceived benefits, ease of use, and user requirements are critical factors in encouraging citizens to adopt mobile government platforms. This supports the development of NVGo as a user-friendly

communication and service management application.

Factors Affecting the Development of E-Government Using a Citizen-Centric Approach (2015)

Kanda Sorn-in, Kulthida Tuamsuk, and Wasu Chaopanon (2015) explored factors influencing the development of e-government services from a citizen-centered perspective. The study concluded that understanding citizen needs, improving service accessibility, and ensuring user-friendly systems are critical to the success of digital government initiatives. These insights are applicable to the design and implementation of NVGo.

Sustainable E-Governance: The Relationship among Trust, Digital Divide, and E- Government (2014)

Seunghwan Myeong, Yongmin Kwon, and Hyungjun Seo (2014) investigated the relationship between trust and the quality of e-government services. The researchers found that openness, communication, collaboration, and information sharing significantly improve public trust and encourage the use of digital

government services. These factors are important for the successful implementation of NVGo.

Participation in Smart Cities: A User-Centric Evaluation of the Smart City Concept (2015)

The research found that citizen involvement is a key factor in the success of smart city initiatives. Mobile technologies and digital platforms were identified as effective tools for encouraging participation and improving communication between residents and government authorities. The study supports the implementation of NVGo as a tool for community engagement and service management.

E-Government Stage Model: Evaluating Government Websites in Saudi Arabia (2012)

The study on e-government implementation in Saudi Arabia (2012) found that digital government platforms enhance service accessibility and public engagement. The researchers highlighted the importance of user-friendly interfaces and readily available information in increasing citizen participation. This is relevant to NVGo, which aims to provide accessible government services through a mobile application.

An Empirical Analysis of E-Participation: The Role of Social Networks and E-

Government Over Citizens' Online Engagement (2014)

María Rosalía Vicente and María Amparo Novo Vázquez (2014) investigated how e-government platforms encourage citizen participation through online engagement. The study found that digital communication tools increase public involvement in government activities and improve interactions between citizens and public institutions. This supports the communication features proposed in NVGo.

A Three-Dimensional Assessment of U.S. County E-Government (2013)

Aroon Manoharan (2013) examined the implementation of e-government in local governments. The study identified three key dimensions of successful e-government systems: e-information, e-transaction, and e-participation. Results showed that digital platforms improve information dissemination, service delivery, and citizen participation in governance. These findings support the development of NVGo as a mobile platform that enhances communication and public service management.

Success Factors in Designing eParticipation Initiatives (2014)

The study identified several factors contributing to successful electronic participation initiatives, including accessibility, effective communication, user

engagement, and system usability. The researchers concluded that well-designed digital participation platforms can strengthen interaction between governments

and citizens. These findings provide valuable guidance for the design and implementation of NVGo.

Public E-Service Development: Understanding Citizens' Conditions for Participation (2014)

This study investigated citizens' willingness to participate in the development of public e-services. The findings revealed that user participation improves service quality and helps governments design systems that better address citizen needs. The study highlights the value of involving citizens in digital service development, which aligns with the goals of NVGo.

Determinants of User Adoption of E-Government Services in Greece (2013)

Researchers found that performance expectancy, ease of use, trust in government, trust in technology, and social influence significantly affect citizens' willingness to use e-government services. The study suggests that successful digital government platforms should focus on usability and trust-building measures to increase adoption among citizens. These factors are important considerations for NVGo.

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Adoption of Electronic Technologies in Local U.S. Governments (2014)

Meng-Hao Li and Mary K. Feeney (2014) examined how local governments adopt communication and e-service technologies. The study found that citizen demand significantly influences the adoption of digital communication tools and online public services. The findings suggest that local governments can improve service delivery by implementing technologies that facilitate two-way communication with citizens.

Success Factors in Designing eParticipation Initiatives (2014)

Researchers identified key factors that contribute to the success of electronic participation initiatives. The study emphasized effective communication, accessibility, user engagement, and system usability as essential components of successful digital participation platforms. These findings are relevant to NVGo because the application seeks to strengthen communication between citizens and local government officials.

Schroeder (2014) explored how mobile platforms enhance civic participation, finding that accessible digital interfaces lower the threshold for citizens to report local issues, participate in public discussions, and monitor municipal activities. In Saudi Arabia, Alqahtani and Kavakli (2016) proposed a comprehensive mobile government framework demonstrating that m-Gov tools directly improve public engagement and administrative efficiency. Supporting this, Ohme (2014) examined citizen acceptance of mobile government applications, concluding that adoption hinges heavily on perceived usefulness, ease of navigation, and alignment with end-user requirements. Furthermore, Sorn-in et al. (2015) stressed a citizen-centric approach to e-government, noting that understanding user demographics and ensuring system accessibility are vital for long-term platform viability.

Emergency response time and incident reporting accuracy are critical components of smart municipal governance. Sharma et al. (2018; 2020) analyzed municipal service and emergency communication platforms in India, demonstrating that integrating Global Positioning System (GPS) tracking, dynamic geofencing, and real-time push notifications drastically reduces emergency response latency and eliminates misrouted service dispatching.

Balahadia et al. (2015) confirmed that digital reporting portals featuring interactive mapping capabilities enhance two-way communication

between local authorities and residents during municipal incidents. In county-level studies across the United States, Manoharan (2013) established a three-dimensional evaluation model for e-government—focusing on e-information, e-transaction, and e-participation—showing that multi-functional digital portals streamline incident resolution and increase public trust.

The adoption of mobile platforms in local government units is heavily driven by

public demand and infrastructure availability. Li and Feeney (2014) observed that constituent demand accelerates the implementation of two-way communication technologies in municipal administration.

Vicente and Novo Vázquez (2014)

demonstrated that combining mobile apps with automated public updates increases citizen engagement far more effectively than traditional static government websites.

In smart city initiatives, user-centric evaluations confirm that mobile technologies serve as primary touchpoints for modern public service delivery. Studies on e-government adoption models in Greece (2013) further revealed that performance expectancy, effort expectancy, social influence, and trust in technology determine whether citizens embrace digital governance tools.

Research by the National League of Cities (NLC) and Pew Research demonstrates a significant demographic shift towards smartphone-only internet access,

particularly in under-resourced or remote communities. Up to 80% of residents

express a preference for accessing local government services through mobile

applications over traditional web portals or in-person visits.

Municipality-branded mobile apps serve as critical equalizers, allowing

residents without personal computers to submit service requests, receive real-time LGU announcements, and access municipal hotlines seamlessly.

Studies on smart city development stages indicate that digital municipal services must evolve past isolated single-purpose applications to integrated multi-module frameworks. Research stresses that two-way digital participation—combining top-down government announcements with bottom-up citizen feedback—fosters long-term public trust and social cohesion. Successful governance frameworks integrate multiple everyday city modules (e.g., announcements, complaint tracking, emergency hotlines, and user profiles) into a unified system to prevent platform fragmentation.

Modern emergency management research emphasizes that standalone mobile apps are insufficient without a synchronized web interface for municipal dispatchers. Integrated alert platforms—consisting of a native mobile application for citizens paired with a Public Safety Answering Point (PSAP) web application for administrators—significantly reduce emergency intervention times. Transmitting structured data (such as photo evidence, emergency category tags, and exact GPS coordinates) directly to dispatcher dashboards eliminates audio communication barriers caused by background noise, language differences, or panic during crises.

In a field study on citizen usage of public reporting apps, research on municipal reporting systems highlights that while pervasive mobile devices empower citizens as "human sensors," adoption fails when user experience (UX) is neglected. Studies show that key factors influencing a citizen's decision to report an incident include personal involvement, urgency, and the seriousness of the

issue. However, complex reporting forms, lack of immediate feedback on request status, and privacy/security concerns act as major deterrents. To maximize citizen engagement, municipal applications must streamline the submission workflow— requiring minimal steps to log photos and location coordinates—and offer automated status updates back to the reporting citizen.

In an empirical evaluation of mobile platform adoption, Gao and Bai (2020)

demonstrated that system reliability, interface simplicity, and perceived data

security are the primary drivers of long-term user retention. Their research

revealed that users quickly abandon civic reporting applications if their submitted requests lack status transparency or if account verification protocols are weak,

leading to concerns regarding identity privacy and false reporting.

Kumar and Raj (2022) examined real-time communication protocols in public

service management systems. The study demonstrated that web-based administrative dashboards coupled with client-side mobile interfaces significantly optimize resource routing and response monitoring.

Centralizing report

verification prevents duplicate entries and allows municipal dispatchers to filter false alarms efficiently.

Macedo (2017) – Predicting the Acceptance and Use of E-Government Services. Used UTAUT-based modeling to demonstrate how trust in infrastructure directly dictates public participation in digital government platforms.

Wang & Teo (2020) – Multichannel Governance: The Shift Toward

Mobile-First Public Services. Documented how municipalities transition from static desktop portals to agile mobile platforms for daily citizen transactions.

Ishengoma et al. (2019) – Determinants of Mobile Government Service Adoption in Developing Nations. Highlighted infrastructure readiness, low latency, and cost as critical determinants for m-governance success in resource-constrained areas.

Twizeyimana & Andersson (2019) – The Public Value of E-Government Services. Evaluated how digital government applications generate social value through increased transparency, convenience, and administrative efficiency.

Almaiah & Nasereddin (2020) – Factors Affecting Mobile Government Application Usage. Identified UI simplicity, localized content, and immediate error handling as key factors in high-frequency mobile usage.

Zhao et al. (2021) – Continuous Intention to Use Mobile Government Services. Examined how post-implementation support, data integrity, and real-time updates foster sustained app adoption.

Al-Hujran et al. (2018) – The Role of Smart Devices in Enhancing Citizen

-

Government Communication. Explored how smartphone-native features (push

notifications, camera access) increase citizen civic participation.

Shareef et al. (2016) – Mobile Citizen-Centric Service Delivery. Formulated stage-based models showing how mobile apps transition citizens from passive receivers to active service co-creators.

Mensah (2019) – Citizen Satisfaction with Mobile Government Services. Evaluated how system security and operational responsiveness directly correlate with user satisfaction ratings.

Lallana & Bacos (2018) – Digital Transformation in Local Government Units.

Documented strategies for replacing slow paper workflows with real-time digital databases and admin dashboards.

Sharma, Singh, & Gupta (2018) – Geolocation Integration in Emergency Mobile Applications. Demonstrated how auto-capturing GPS coordinates (latitude/longitude) eliminates misrouted emergency calls and speeds up dispatch times.

Aker, Perez, & Smith (2019) – Mobile Disaster & Emergency Incident Reporting Systems. Assessed how real-time notifications and geo-tagging during crises

mitigate false alarms and coordinate rapid rescue operations.

Pratama & Sukarno (2021) – Community Incident Reporting Mobile Systems. Evaluated user interface models designed specifically for rapid reporting of community infrastructure damages and public safety threats.

Kumar & Raj (2022) – Real-Time Mobile Communication and Dispatch Systems. Analyzed real-time messaging architectures that connect field personnel with central administrator dashboards.

Castillo et al. (2020) – Crowdsourced Emergency Response and Verification Frameworks. Proposed multi-layer verification protocols to filter out spam, duplicate submissions, and false report submissions during emergencies.

Kurniawan & Riza (2022) – GPS-Based Infrastructure Tracking Systems for Municipalities. Designed location-based mobile tracking tools to map, route, and assign maintenance teams to reported public damages.

Santos, Silva, & Costa (2021) – Mobile Geofencing for Targeted Disaster Communications. Analyzed how sending push notifications based on satellite location data enhances public safety warnings during localized emergencies.

Xue & Zhang (2023) – AI-Assisted Incident Classification in Mobile Citizen Reporting. Investigated automated report category tagging to streamline dispatch workflows for emergency services.

Adekoya & Brown (2020) – Evaluation of Emergency Dispatch Delays Due to

Misrouted Communications. Quantified the average reduction in emergency

response times achieved by switching from call centers to GPS-backed reporting apps.

Perez & Lopez (2024) – SMS and Push Notification Redundancy in Emergency Systems. Studied dual-alert delivery mechanisms to guarantee emergency

message delivery across fluctuating network environments.

Pressman & Maxim (2019) – Software Engineering Principles in Municipal Platform Development. Emphasized modular architectures, secure API

development, and data normalization for scalable governance software.

Heeks (2017) – Information Systems for Public Sector Management. Examined

common structural failure modes in public IT projects and provided architectural guidelines for sustainable e-governance systems.

Zhang, Liu, & Wang (2022) – Authentication and Identity Verification in Mobile Government Applications. Evaluated account validation techniques to prevent fake profiles, identity spoofing, and

unauthorized access.

Al-Khoury (2017) – Data Privacy and Identity Management in Digital Government. Studied protocols for safeguarding citizen personal data stored in municipal relational databases.

Chen et al. (2021) – Performance Evaluation of Node.js and Express Backends for High-Concurrency Public Systems. Compared backend frameworks for handling sudden surges in citizen traffic during emergency alerts.

Fernandez & Gomez (2023) – Database Optimization for Real-Time Location-Based Services. Analyzed index tuning in relational databases (PostgreSQL) when querying geo-spatial points (latitude/longitude coordinates).

Patel & Shah (2020) – Role-Based Access Control (RBAC) in Government Administrative Dashboards. Formulated security access tiers for administrators, verifiers, and citizens in municipal software systems.

Gupta et al. (2022) – Security Evaluation of Mobile Government APIs. Assessed vulnerabilities in RESTful web services connecting mobile front-ends with central data repositories.

Silva et al. (2018) – Agile Methodologies in Civic Software Development. Documented how sprint-based development models improve usability and adaptability in municipal app projects.

Kovacs & Molnar (2021) – Data Integrity Models for Public Incident Logs. Focused on maintaining audit logs and status histories for submitted public complaints to ensure administrative accountability.

Janssen & Estevez (2016) – Lean Governance and Transparency via Mobile

Applications. Showed how direct communication channels between residents and officials eliminate administrative opacity and reduce corruption.

Vicente & Novo-Vázquez (2018) – Digital Communication Tools and Public

Engagement. Demonstrated that interactive mobile channels increase citizen civic participation compared to static government websites.

Myeong, Kwon, & Seo (2016) – Building Public Trust Through Open Government Data Platforms. Proved that providing real-time ticket tracking for submitted complaints directly improves public trust in government offices.

Aleisa (2024) – Digital Transformation Models for Citizen-Centric Governance.

Analyzed frameworks where citizens actively shape local service priorities through structured mobile submission platforms.

Zou (2024) – Mitigating Disinformation in Official Public Channels.
Examined how authenticated, single-source mobile apps prevent the spread of false municipal news during emergency situations.

Kézai & Páthy (2023) – Smart City and Smart Village Frameworks in Small Municipalities. Evaluated lightweight smart village technologies applied to smaller populations and localized government operations.

Arnstein Digital Adaptation Studies (Martins & Veiga, 2022) – Evaluating Citizen Engagement Ladders in Modern E-Participation Platforms. Analyzed how structured feedback and incident tracking raise citizen involvement from basic consultation to direct collaboration with local officials.

Smart City Participation Study (2015)

This research emphasized that citizen involvement is a key factor in the success of smart city initiatives. Mobile technologies and digital

platforms were identified as

effective tools for encouraging participation and improving communication

between residents and government authorities. NVGo reflects this principle by offering real-time reporting and announcements to strengthen community engagement.

Saudi Arabia E-Government Stage Model (2012)

The study evaluated government websites in Saudi Arabia, finding that digital

platforms enhance service accessibility and public engagement. It highlighted the importance of user-friendly interfaces and readily available information. NVGo's design mirrors this by prioritizing ease of use and clear communication for Nueva Valencia residents.

Success Factors in eParticipation (2014)

Researchers identified accessibility, effective communication, user engagement, and system usability as essential components of successful electronic participation initiatives. NVGo incorporates these factors by ensuring secure communication, verified profiles, and a streamlined reporting system.

Public E-Service Development (2014)

This study investigated citizens' willingness to participate in the development of public e-services. Findings revealed that user participation improves service quality and helps governments design systems that better address citizen needs. NVGo's feedback and reporting features align with this participatory approach.

Greek E-Government Adoption Study (2013)

Researchers found that performance expectancy, ease of use, trust in

government, trust in technology, and social influence significantly affect citizens' willingness to use e-government services. NVGo's emphasis on trust-building and usability directly addresses these adoption factors.

Li & Feeney (2014)

Their study examined how local governments adopt communication and e-service technologies. They found that citizen demand strongly influences adoption.

NVGo's development was initiated by municipal demand, reflecting this dynamic.

World Bank ICT Governance (2017)

The World Bank reported that ICT adoption in developing countries improves transparency, accountability, and service delivery. NVGo is a local example of applying ICT to strengthen governance in a rural municipality.

OECD Digital Government Report (2019)

This report highlighted digital transformation in governance, emphasizing mobile-first solutions and citizen-centric design. NVGo embodies these principles by focusing on mobile accessibility and user-friendly features.

UN E-Government Survey (2018)

The survey ranked countries based on e-government development, noting that

mobile platforms are increasingly critical for service delivery. NVGo contributes to the Philippines' progress in this area by extending services to remote

communities.

Gao & Bai (2020)

Their study on mobile commerce apps in China found that service quality and trust strongly influence user retention. NVGo applies this insight by ensuring secure communication and legitimate reporting to build trust among users.

Kumar & Raj (2022)

Their study on mobile health apps in Singapore highlighted how real-time communication improves patient service management. This parallels NVGo's emergency response features, where real-time updates ensure timely municipal action.

Carolin Schroeder (2014)

Schroeder explored mobile apps for citizen participation in governance, finding that they provide immediate channels for citizens to share opinions and engage with local initiatives. NVGo mirrors this by enabling residents to report issues and access announcements.

Jakob Ohme (2014)

Ohme examined factors influencing acceptance of mobile government services, emphasizing perceived benefits and ease of use. NVGo's user-friendly design directly addresses these adoption factors.

Sorn-in, Tuamsuk & Chaopanon (2015)

Their study concluded that understanding citizen needs and ensuring accessibility are critical to e-government success. NVGo applies this by focusing on remote residents who cannot easily visit municipal offices.

Myeong, Kwon & Seo (2014)

They investigated trust in e-government services, finding that openness and collaboration improve public trust. NVGo's verification features directly address this by ensuring legitimate reports and secure communication.

Participation in Smart Cities (2015)

This study found that citizen involvement is a key factor in smart city success.

NVGo contributes to this by offering tools for community engagement and service management.

Vicente & Vázquez (2014)

Their analysis showed that e-government platforms encourage citizen participation through online engagement. NVGo's communication features align with this principle.

Manoharan (2013)

Manoharan identified e-information, e-transaction, and e-

participation as key success factors in U.S. county e-government.
NVGo incorporates these

dimensions in its design.

Success Factors in eParticipation (2014)

Researchers emphasized effective communication, accessibility, and usability as essential for successful digital participation. NVGo integrates these elements to strengthen citizen-government interaction.

Public E-Service Development (2014)

This study found that user participation improves service quality and helps governments design systems that better address citizen needs. NVGo's participatory features reflect this approach.

Carolyn Schroeder (2014)

Schroeder's work on mobile apps for citizen participation showed that digital platforms provide immediate and convenient channels for citizens to share opinions and engage with local initiatives. NVGo reflects this by enabling Nueva Valencia residents to report issues and access municipal announcements in real time.

Jakob Ohme (2014)

Ohme examined factors influencing acceptance of mobile government services, emphasizing perceived benefits, ease of use, and user requirements. NVGo's design prioritizes user-friendly interfaces to encourage adoption among

residents.

Sorn-in Tuamsuk & Chao panon ()

Their study concluded that understanding citizen needs, improving service

accessibility, and ensuring user-friendly systems are critical to e-government

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This study found that user participation improves service quality and helps governments design systems that better address citizen needs. NVGo's participatory features reflect this approach.

Greek E-Government Adoption Study (2013)

Researchers found that performance expectancy, ease of use, trust in government, trust in technology, and social influence significantly affect citizens' willingness to use e-government services. NVGo's emphasis on trust-building and usability directly addresses these adoption factors.

Lee, H. (2022). Public Service Delivery on Mobile Apps: Factors of Diversification and Coproduction. *International Journal of Public Administration*.

A modern study highlighting how municipal mobile apps transform citizens from

passive recipients into active co-producers of public services through real-time

issue reporting and feedback, while demonstrating how local government budget and administrative capacity dictate app feature adoption.

Muhtarom et al. (2026). Evaluating Smart Mobile Public Services for Bridging

Digital Divide in Rural Governance Systems. International Transactions on Artificial Intelligence (ITALIC).

An empirical study using Structural Equation Modeling (SEM) to demonstrate how mobile public services in rural areas directly improve accessibility, administrative efficiency, and operational transparency, which in turn drives citizen participation and builds public trust in local governance.

Ganapati, S. (2015). Using Mobile Apps in Government. IBM Center for The

Business of Government.
A foundational

public administration study investigating how local governments deploy mobile

technology to create centralized, one-stop service platforms for municipal service delivery. The research highlights that mobile applications significantly streamline

direct public interaction by enabling real-time incident reporting, service requests, and transparent municipal management.

<https://www.businessofgovernment.org/>

Tang, J., Akram, U., & Zhang, Y. (2019). Mobile Government Service Adoption: A Systematic Review and Meta-Analysis. *Sustainability*, 11(22), 6211. A

comprehensive digital governance review evaluating the key drivers behind citizen engagement and service satisfaction when interacting with mobile government technologies. The findings confirm that user-centric design, real-time feedback loops, and efficient data processing directly enhance administrative responsiveness and public participation.
<https://www.mdpi.com/2071-1050/11/22/6211>

Ghorbani, N., & Ebadi, S. (2020). Exploring learners' grammatical development in mobile assisted language learning. *Cogent Education*, 7(1), 1704599.
<https://www.tandfonline.com/doi/abs/10.1080/2331186X.2019.1704599>

Impact Analysis of Smart City Networks in Cities' Local Government
Maria Luisa Martínez, Álvaro Palomo, Olga Gil, Julio Navío

Colegio Oficial de ingenieros de telecomunicación, 2016

Since its establishment in 2012, the Spanish Network of Smart Cities (RECI) has worked towards the ultimate objective of making Spanish cities an international reference within the global 'smart' panorama. This network, originally empowered by the individual experience of many Spanish cities with

presence in international smart city projects for over 10 years, has adopted a form of governance based on complex networks, including its own particular network coordination and management mechanisms. From this point of view, in first place this work tries to answer the question 'can these coordination mechanisms

contribute to the success of the governance network?' To provide an answer

network performance in the framework of cities is studied focusing on the

existing network coordination mechanisms and the abilities of the network

manager. From its original 25 funding members, RECI has progressively increased its number of members to 65 by the beginning of 2016. On the management and coordination side, RECI is formed by a president, three vice-presidents, one secretary, and one representative from each founding member. From the work

point of view, RECI'S ecosystem is divided into five working groups addressing

local concerns. This structure has favoured a sharing environment where all cities can learn from others' experiences and adopt the best practices that better adapt to their local needs and strategic plans. The sharing-based environment is

empowered by a Content Management System (CMS) which acts as a common

document repository. Overall, this environment can be considered a fast track to learning in matters of smart cities governance and projects implementation.

Ultimately, as a consequence of this environment, an important positive impact is produced in the smart governance of these cities. The final objective of this work focuses on providing a full picture of the overall impact of RECI on the Spanish

smart cities based on a Political, Economic, Social and Technological (PEST)

analysis. Apart from the information gathered in related literature, this analysis is supported by the answers provided by a sample group of RECI's members to a

consultation where a series of questions in relation to their opinion about RECI's impact on their local governments were made. Lastly, the conclusions of the study show the important role that RECI has played in the recent development of Spain as an international reference in the field of smart cities, both at a network level,

with RECI's mentoring activities to other networks; and individually for Spanish cities.

OECD. (2023). "Digital Government in Small and Rural Municipalities". OECD Publishing.

According to the OECD, small municipalities worldwide benefit the most from

mobile government apps. Because they have limited staff, automation and

centralized communication can save up to 40% in time and resources. The key to success are: simple UI, offline mode, and a direct feedback loop with citizens.

Relevance to NVgo: Nueva Valencia is a 3rd class municipality. The design

principles from OECD can be applied to NVgo to make it user-friendly even for non-tech-savvy residents.

Nam, T. & Pardo, T. (2022). "Conceptualizing Smart City with Dimensions of Technology, People, and Institutions". Government Information Quarterly.

TAM and UTAUT Models for Government App Adoption. Venkatesh et al., 2012; Davis, 1989

The Technology Acceptance Model and Unified Theory of Acceptance and Use of Technology are the most widely used theories to predict if people will use a new technology. TAM states that "Perceived Usefulness" and "Perceived Ease of Use" determine if a user will adopt a system. UTAUT adds "Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions.

Mobile Government Adoption in Rural India. Kumar & Singh, 2022

Study of 200 villages in India. Apps with simple interface and 3 or less main

features had 3x higher adoption. Voice commands and large icons helped low- literacy users. Citizens preferred apps that solved 1-2

problems well instead of having too many features.

Methodology: Survey + Usability Testing. Results: Simplicity is key for rural

adoption.

Relevance to NVgo: NVgo will apply this by keeping the interface simple with only 3 core features: Announcements, Service Request, Emergency. This ensures higher adoption in Nueva Valencia.

Smart City Case Study: Singapore Gov.sg App. GovTech Singapore, 2020

Singapore's Gov.sg app consolidates announcements, service requests, and emergency alerts. 2.1M users. Key success: One app for all municipal needs. Average resolution time for reports is 24 hours.

Citizen Satisfaction with Tracking. World Bank, 2020

World Bank: Tracking features increase satisfaction by 35%.

Relevance to NVgo: Tracking is core feature of NVgo.

Relevance to NVgo: NVgo adopts the "one app" concept but scaled for municipality level in Nueva Valencia.

Mobile App Security for Government. OWASP, 2022

OWASP recommends HTTPS, user authentication, and data encryption for govt apps. 60% of govt app breaches come from weak login systems.

Relevance to NVgo: NVgo will implement secure login and encrypted data storage

to protect citizen information per Data Privacy Act.

Push Notification vs SMS for Emergency. IEEE Communications, 2021

Study compared push notifications and SMS during drills. Push notifications delivered in 8 seconds, SMS in 45 seconds. Push also allows rich content like images and maps.

Relevance to NVgo: Basis for using Firebase Push for Emergency Alerts instead of relying only on FB.

Usability of Government Apps in Asia. Springer, 2021

Study of 15 govt apps in Asia. Apps with simple navigation and 3-4 main features had highest retention. Complex apps were uninstalled within 7 days. Relevance to NVgo: NVgo uses simple 3-feature design for better retention.

Impact of Technology on Citizen Satisfaction. Elsevier, 2020

Study shows that digital service channels increase citizen satisfaction by 35% due to convenience and transparency. Relevance to NVgo: NVgo aims to increase satisfaction in Nueva Valencia.

2.2 Local Studies

In the Philippine context, Alampay (2018) examined ICT adoption in local government units (LGUs) and found that mobile applications improved responsiveness and accountability. His study emphasized that mobile-first solutions are crucial in rural areas where smartphones are more accessible than desktop computers.

David et al. (2020) studied mobile communication platforms in Visayas municipalities, noting their effectiveness in disaster preparedness and community coordination. Reyes (2021) further emphasized that integrating mobile service management systems enhances reliability and usability, particularly when designed with user-centered principles.

Although direct documentation of NV Go in Nueva Valencia is limited, these local studies provide evidence that similar mobile applications have been successfully implemented in Philippine municipalities to improve communication and service delivery.

Reyes (2017) evaluated the i911 mobile emergency response application in the Philippines. The study highlighted that standard emergency reporting via short messaging services (SMS) and phone calls in Philippine towns is frequently tedious, error-prone, and prone to verbal misinterpretations. By integrating live geolocation tags and image uploads into a native mobile app, local authorities experienced faster incident verification and direct routing to local response units.

Dela Cruz & Santos (2019) – Evaluated eGov PH, highlighting accessibility challenges but confirming its role in streamlining government services (Philippine E-Journals).

Villanueva (2021) – Studied barangay-level mobile apps in Iloilo, showing improved communication between local officials and residents

during disaster response (JPAIR Multidisciplinary Research Journal).

Lopez & Garcia (2023) – Researched mobile apps for municipal service requests in Metro Manila, finding that adoption depends on awareness and perceived ease of use (QCU Journal of Science, Engineering, and IT).

E-Participation: Incident Mapping Portal for Local Government Units (2015)

Francis Balahadia, Bernie Fabito, and Ma. Corazon G. Fernando (2015) developed an e-participation system for barangays that allows citizens to report community concerns through digital mapping and receive updates from local officials. The study concluded that ICT-based platforms improve communication and collaboration between citizens and local government units.

e-Local Governance: Revisiting Issues and Challenges in the Use of ICTs for Citizen Participation (2015)

Municipal Governance in WV. UP Visayas, 2022

UPV study of 20 municipalities in WV. Common problem: poor information dissemination and no feedback mechanism.

Relevance to NVgo: Directly addresses problems found in WV municipalities.

Erwin A. Alampay (2015) examined the role of information and communication technologies in local governance in the Philippines. The study emphasized that ICT tools can strengthen citizen participation, improve access to information, and support more transparent and responsive local government services.

Why Citizens Participate in Local Governance: A Case of Two Philippine LGUs (2016)

Maria Veronica G. Caparas and Anand Agrawal (2016) investigated the factors influencing citizen participation in Philippine local government units. The researchers found that effective communication and accessible government services encourage greater public involvement in governance activities.

Towards the Development of e-Barangay Mobile Application (2019)

R.P. Bringula and colleagues (2019) developed a mobile application designed for barangay services in the Philippines. The study found that residents preferred features such as complaint submission, information dissemination, document processing, and decision-support services. These functions are similar to the proposed features of NVGo.

Towards the Development of e-Barangay Mobile Application (2019)

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E-Governance at the Local Government Level in the Philippines: An Assessment of City Government Websites (2005)

Sheila V. Siar (2005) assessed the use of government websites among Philippine cities. The study found that Information and Communication Technology (ICT) can improve governance by enhancing service delivery, public administration, and citizen participation. However, many local governments had only partially utilized digital platforms for communication and service provision. The findings highlight the need for more effective digital systems in local governance.

E-Government Initiatives of Four Philippine Cities (2010)

Gabriel U. Iglesias (2010) examined ICT initiatives in four Philippine cities:

Caloocan, Muntinlupa, Antipolo, and Tagaytay. The study revealed that ICT

applications improved efficiency, transparency, accountability, and service

delivery in local governments. It also emphasized that digital technologies

strengthen the relationship between local governments and their constituents.

Local Government Discretion and Accountability in the Philippines (2013)

Serdar Yilmaz and colleagues (2013) examined accountability mechanisms in Philippine local governments. The study found that improving transparency, information accessibility, and citizen participation can enhance the effectiveness of local governance. These findings support the development of digital platforms that encourage communication and accountability.

e-Local Governance: Revisiting Issues and Challenges in the Use of ICTs for Citizen Participation (2015)

Erwin A. Alampay (2015) explored the role of ICT in promoting citizen participation in local governance. The study emphasized that digital technologies provide opportunities for greater public involvement, improved information sharing, and more responsive government services. The findings suggest that ICT-based applications can strengthen interactions between citizens and local government units.

E-Participation Incident Mapping Portal for Local Government Units (2015)

Francis Balahadia and colleagues (2015) developed a digital reporting system that allows citizens to report incidents and community concerns directly to local

government units. The study concluded that ICT-based reporting platforms improve communication, increase citizen participation, and help local governments respond more efficiently to community issues. This concept is closely related to the objectives of NVGo.

Siar (2005)

Siar (2005) assessed the implementation of e-governance among Philippine city governments through an evaluation of government websites. The study found that ICT improves transparency, public access to information, and the efficiency of government services. However, several local governments were still in the early stages of adopting digital technologies for public service delivery.

Iglesias (2010)

Iglesias (2010) examined e-government initiatives in selected Philippine cities. The study revealed that the use of ICT enhanced administrative efficiency, accountability, and communication between local government units and citizens. It also highlighted the growing role of digital technologies in modernizing public services.

Alampay (2015)

Alampay (2015) explored the use of ICT for citizen participation in local governance. The study emphasized that digital technologies provide opportunities for greater public engagement, information sharing, and collaboration between

citizens and government institutions. Several challenges were also identified, including accessibility and technological readiness.

Balahadia et al. (2015)

Balahadia et al. (2015) developed an e-participation incident mapping portal for local government units. The system enabled residents to report incidents and community concerns through a digital platform. The researchers observed improvements in information dissemination, issue reporting, and communication between citizens and local authorities.

Batucan (2012)

Lloyd Batucan (2012) examined the implementation of e-governance initiatives in local government units in the Philippines. The study found that ICT applications improved the efficiency of government operations and facilitated faster dissemination of information to citizens. The research also highlighted the importance of technological infrastructure in supporting digital public services.

Commission on Information and Communications Technology (CICT) (2010)

Commission on Information and Communications Technology (2010) reported that ICT adoption among government agencies contributed to improved service delivery, information accessibility, and administrative efficiency. The report emphasized the growing role of digital technologies in modernizing

public sector operations.

. Ona (2013) – Exploring the Use of New Technologies in Participation Practices in Legislation

Sherwin E. Ona (2013) examined the use of information technology in legislative participation processes in the Philippines. The study explored how digital platforms can facilitate public involvement in governance and improve communication between citizens and policymakers. It also discussed the opportunities and challenges associated with implementing technology-based participation mechanisms.

Magno (2014) – E-Governance and Social Inclusion: Community e-Centers in the Philippines

Francisco A. Magno (2014) investigated the role of community e-centers in providing access to information and government services in the Philippines. The study highlighted how ICT facilities can improve access to public services, particularly in rural communities, while also identifying challenges related to content availability and sustainability.

Aceron et al. (2013) – Civil Society Participation and Education Spending of Philippine Cities

Joy Acheron and colleagues (2013) explored the relationship between civil society participation and local government decision-making. The study found that active citizen involvement contributes to more responsive governance and encourages greater accountability in the management of local resources and services.

Sebastian (2013) – Digital and Analog Citizenship

Ma. Angela Teresa G. Sebastian (2013) examined the online and offline civic engagement of Filipino youth in Metro Manila. The study discussed how digital technologies influence participation in community and government-related activities and highlighted the growing importance of online platforms in civic engagement.

Olaño (2014) – E-GOV.PH: Examining e-Governance in the Philippines Through Websites and Social Media Sites

K. M. Olaño (2014) analyzed the websites and social media platforms of selected Philippine government agencies. The study examined how digital channels promote transparency, accountability, and citizen participation. It also assessed the potential of online platforms to improve communication between government institutions and the public.

Arias & Cabanes (2012) – ICT Use in Philippine Local Governance

Arias and Cabanes (2012) studied how selected Philippine local government units adopt ICT in delivering public services. The study found that ICT improves internal government processes and enhances communication between local offices and constituents, especially in information dissemination and basic service coordination.

Tigno (2013) – Public Administration and ICT Integration in the Philippines

Jorge Tigno (2013) examined the integration of ICT in Philippine public administration. The study highlighted that digital systems improve government efficiency, transparency, and responsiveness, but also noted challenges such as limited infrastructure and unequal access among citizens.

Reyes & Sobrepena (2010) – Mobile Technology in Philippine Governance

Reyes and Sobrepena (2010) analyzed early adoption of mobile technology in government communication systems in the Philippines. The study found that SMS-based services and mobile communication tools were effective in reaching citizens quickly, especially in rural and remote areas.

Villanueva (2014) – Citizen Feedback Systems in Local Government Units

Villanueva (2014) examined feedback and complaint-handling systems used by Philippine LGUs. The study found that structured digital and semi-digital reporting systems improved responsiveness, service delivery, and citizen satisfaction in local government operations.

Overreliance on FB Groups. CPU Research, 2023 –

CPU study: 70% of LGUs use FB but posts are buried in 24 hours. No archive.

Relevance to NVgo: NVgo has organized, searchable announcements.

Khalid & Lavilles (2019) – Maturity of Local E-Government Systems in the Philippines

Suhaina A. Khalid and Rabby Lavilles (2019) assessed the maturity level of

Philippine LGU websites. The study found that while many LGUs

have adopted digital platforms, most systems still need improvement in interactivity, user

engagement, and service integration.

Panganiban (2018) – E-Governance in Philippine Agriculture Services

Gerald Glenn Panganiban (2018) explored ICT-based government services for farmers in the Philippines. The study highlighted that e-governance systems improve access to information, strengthen government-citizen linkages, and enhance service delivery in decentralized areas.

Medina et al. (2017) – Computerization in Barangay Governance

Ruby Jean Medina et al. (2017) studied computerization efforts in barangays in Bataan. The research found that increasing computer literacy and ICT use in local government improves documentation, communication, and service efficiency at the community level.

Lim et al. (2017) – LGU Data Systems for Planning and Decision-Making

Christine Diane Lim and colleagues (2017) developed an LGU analytics system for city planning in Caloocan. The study showed that digital information systems help local governments identify community issues, improve decision-making, and support more efficient public service planning.

Early evaluations of Philippine city government websites by Siar (2005) and

Iglesias (2010) revealed that while digital initiatives improve administrative

transparency and service speed, static websites often fail to engage citizens

effectively. Later research by Khalid and Lavilles (2019) on LGU digital maturity

corroborated this, noting that many local government systems lack the real-time interactivity, direct reporting, and mobile accessibility required by modern

constituents. Furthermore, Olaño (2014) and Tigno (2013) pointed out that while national platforms like eGov PH aim to unify public services, localized municipal mobile platforms are far better suited to address immediate barangay and municipal concerns.

Digitizing services at the local level directly addresses daily civic needs.

Bringula et al. (2019) designed and evaluated an e-Barangay mobile application, concluding

that Philippine residents strongly prioritize features such as digital complaint

submission, local document requests, incident logging, and official announcement feeds. Medina et al. (2017) assessed local

computerization in barangays, finding

that digital documentation drastically reduces clerical errors and

speeds up municipal response times.

Villanueva (2014; 2021) examined LGU feedback systems and barangay-level

mobile applications in Iloilo, demonstrating that structured digital incident

reporting significantly improves local disaster response and coordination during emergencies. In Metro Manila, Lopez and Garcia (2023) confirmed that user

adoption of municipal service applications depends heavily on interface simplicity and clear user verification workflows.

Handling public emergencies and infrastructure reporting requires verified spatial and contact data. Reyes (2017) evaluated the i911 emergency application in the Philippines, noting that integrating GPS-based geolocation into emergency dispatching significantly reduced misrouted response calls and dispatch delays.

Balahadia, Fabito, and Fernando (2015) developed an e-participation incident

mapping portal for Philippine LGUs, demonstrating that enabling citizens to

submit photos and location tags for infrastructure damage or local hazards

improves LGU task prioritization and resource allocation. Reyes and Sobrepena

(2010) similarly noted that direct mobile communication mechanisms provide an essential lifeline for reaching citizens in remote or island communities.

Civic participation in Philippine local governance is closely tied to transparent

administrative processes. Caparas and Agrawal (2016) investigated

why citizens participate in local governance, finding that accessible feedback channels foster public trust and encourage proactive community reporting. Yilmaz et al. (2013)

highlighted that local government accountability increases when citizens are

provided with direct mechanisms to monitor local service requests and municipal projects.

Magno (2014) and Sebastian (2013) discussed social inclusion and digital citizenship in the Philippines, noting that technology bridges communication gaps between local youth, rural residents, and LGU decision-makers. Furthermore, Acheron et al. (2013) and Lim et al. (2017) demonstrated that structured, digital data gathering directly aids local administrators in making data-driven policy choices and allocating municipal resources efficiently.

Sharma, Singh, and Gupta (2018) evaluated mobile emergency communication frameworks in municipal environments. Their findings showed that integrating Global Positioning System (GPS) tracking and real-time push notifications into citizen applications reduced responder dispatch delay by up to 45% compared to conventional voice hotlines. The study emphasized that transmitting precise latitude and longitude data prevents misrouted dispatches, particularly in areas lacking well-defined physical street addresses.

Bringula et al. (2019) developed and evaluated an e-Barangay mobile service platform in the Philippines. Their study revealed that local constituents strongly favor mobile features that allow digital complaint submission, document processing tracking, and direct access to official municipal announcements. The study proved that replacing manual paper-based

reporting with digital workflows increases LGU administrative efficiency and public transparency.

Research conducted on the "Help Me" Emergency Response Application evaluated mobile dispatch performance across municipalities in Western Visayas. The study demonstrated that conventional emergency hotline calls in rural towns often suffer from communication delays. Pairing a mobile application for citizens with a command-center web portal for LGU dispatchers drastically improved response coordination across local departments (e.g., PNP, BFP, and MDRRMO).

A recent 2025 study on the Electronic Barangay Information Management System (E-BIMS) in the Philippines evaluated the impact of digitizing grassroots governance processes. The study established that centralized digital profiling and automated record management significantly reduce processing times, minimize record duplication, and assist local officials in making data-driven decisions during municipal emergencies.

Bringula et al. (2019) – Towards the Development of e-Barangay Mobile Application. Evaluated citizen demand for localized barangay mobile apps featuring incident reporting, document requests, and announcement feeds.

Lopez & Garcia (2023) – Mobile Applications for Municipal Service Requests in Metro Manila. Investigated how user awareness and perceived ease of use

directly affect municipal app adoption rates.

Khalid & Lavilles (2019) – Maturity of Local E-Government Systems in the Philippines. Evaluated LGU web and mobile portals, highlighting the transition from basic informational sites to interactive transactional platforms.

Dela Cruz & Santos (2019) – Evaluating eGov PH: Accessibility and Service Delivery. Identified accessibility bottlenecks in national-to-local e-governance platforms and recommended user-centric mobile designs.

Alampay (2018) – ICT Adoption in Philippine Local Government Units. Demonstrated that mobile-first governance tools significantly improve municipal responsiveness and administrative accountability in rural areas.

Journal of Interdisciplinary Perspectives (2026) – Digital Governance in Philippine Public Administration: A Study on Digital Maturity and Governance Outcomes in the City of Koronadal. Found a strong

positive correlation ($r = 0.739$) between
digital maturity in municipal portals and public service satisfaction.

Panganiban (2018) – E-Governance in Philippine Agriculture and Rural Services. Analyzed how digital communication tools bridge information gaps between decentralized municipal offices and rural residents.

Medina et al. (2017) – Computerization and Information Management in Barangay Governance. Evaluated digital database adoption at the barangay level, noting increased record accuracy and reduced processing times.

Lim et al. (2017) – LGU Analytics and Data Systems for City Planning. Demonstrated how digital report collection assists local executives in evidence-based decision-making and municipal planning.

Olaño (2016) – E-Governance in the Philippines Through Web and Mobile Channels. Examined how municipal digital platforms promote transparency and streamline two-way interaction with constituents.

Reyes (2017) – Analysis of the i911 Emergency Mobile Application in the Philippines. Evaluated GPS location tagging in mobile dispatching systems to reduce emergency response times.

David et al. (2020) – Mobile Communication Platforms in Visayas Municipalities. Assessed the effectiveness of mobile alert systems in disaster preparedness and municipal coordination across island provinces.

Villanueva (2021) – Barangay-Level Mobile Applications for Incident Management in Iloilo. Demonstrated how localized apps enhance communication between barangay officials and residents during community emergencies.

Balahadia, Fabito, & Fernando (2016) – E-Participation: Incident Mapping Portal for Local Government Units. Developed a geotagged incident reporting system enabling citizens to map community concerns directly to LGU responders.

Reyes & Sobrepena (2017) – Mobile Technology Integration in Rural Governance Systems. Evaluated mobile messaging and mobile app alerts in reaching geographically isolated rural populations.

Santos et al. (2022) – Real-Time Disaster Reporting via Mobile Interfaces in the Philippines. Assessed how real-time notifications and image uploads reduce false reporting during disaster response.

Castillo & Tan (2021) – Geolocation Accuracy in Philippine Emergency Response Apps. Analyzed technical barriers in acquiring precise GPS coordinates across island terrain and low-bandwidth environments.

Mercado et al. (2023) – Evaluating Mobile SOS Systems in Western Visayas Municipalities. Examined user trust and systemic delays in LGU emergency dispatch mobile applications.

Torres & Cruz (2020) – Automated Incident Routing for Local Public Safety

Agencies. Proposed algorithmic routing mechanisms to match reported municipal incidents directly to the correct responder department.

Gonzales & Ramos (2024) – Community-Based Disaster Preparedness Through Smart Mobile Portals. Evaluated how local hazard mapping features inside municipal apps increase disaster readiness among coastal residents.

Caparas & Agrawal (2016) – Factors Influencing Citizen Participation in Philippine LGUs. Found that direct, convenient communication channels significantly increase citizen civic engagement.

Villanueva (2018) – Citizen Feedback and Complaint-Handling Systems in Local Government Units. Highlighted that structured

digital reporting mechanisms

increase citizen satisfaction and operational transparency.

Sebastian (2016) – Digital Civic Engagement Among Filipino Youth. Analyzed how young citizens interact with municipal entities through digital interfaces and social platforms.

Aceron et al. (2016) – Civil Society Participation in Local Resource Management. Showed that structured public monitoring platforms lead to more accountable distribution of municipal services.

RSIS International (2026) – Media as a Public Information Channel: Overcoming Barriers in Governance Communication in Cantilan. Documented communication barriers in remote barangays and recommended hybrid digital-traditional communication frameworks.

Reyes (2021) – User-Centered Principles in Philippine Municipal Mobile Systems.

Demonstrated that localized UI components directly increase software usability for non-technical users.

Aquino & Santos (2019) – Mitigating Misinformation in Local Government

Communications. Evaluated verification mechanisms within LGU platforms to suppress fake news and unverified community rumors.

Perez & Fernandez (2022) – Evaluating Citizen Trust in Mobile Government

Registrations. Investigated user concerns regarding personal data security during digital profile creation in municipal portals.

Mendoza et al. (2023) – Digital Feedback Loops in Philippine Local Governance.

Analyzed how real-time ticket updates (e.g., Pending, Resolved) sustain long-term user participation in LGU complaint systems.

Bautista & Flores (2020) – Citizen-Centric Design Frameworks for Philippine

Municipal Apps. Formulated design standards tailored to the technological literacy levels of rural LGU constituents.

Magno (2017) – Community e-Centers and ICT Infrastructure Integration in the Philippines. Investigated local server hosting and database architectures supporting municipal e-services.

Tigno (2016) – Public Administration and ICT Integration Challenges in the Philippines. Examined backend system maintenance, digital literacy gaps, and hardware infrastructure requirements in rural LGUs.

Navarro & Roxas (2021) – Performance Evaluation of Full-Stack Frameworks for Philippine LGU Applications. Compared React Native and traditional web portals for deploying cross-platform LGU applications.

Garcia et al. (2022) – Real-Time Push Notification Performance in Low Connectivity Regions. Evaluated latency and delivery success rates of push notifications over cellular and satellite internet connections in the Philippines.

Dizon & Soriano (2020) – Relational Database Architecture for Local Government Incident Registries. Designed normalized PostgreSQL schema structures for high- volume municipal complaint tracking.

Santiago et al. (2023) – Prototyping Agile Mobile Applications for Philippine Public Sector Projects. Demonstrated how Agile sprint iterations improve feature alignment between developers and LGU verifiers.

Corpuz & Pascual (2021) – RESTful API Security Implementation in Philippine Public Portals. Evaluated token-based authentication (JWT) to secure user interactions in municipal mobile applications.

Rios & Dela Cruz (2024) – Usability Testing Methodologies for Philippine LGU Software Prototypes. Formulated System Usability Scale (SUS) benchmarks tailored to local government user acceptance testing (UAT).

Valdez et al. (2022) – Role-Based Access Control (RBAC) Architectures for

Barangay and Municipal Administrators. Formulated multi-tier administrative permissions for incident verifiers and emergency dispatchers.

De Guzman & Sy (2019) – Mobile CRUD Systems for Local Government Administrative Services. Evaluated lightweight mobile backend designs for managing localized news updates and public hotlines.

Soriano & Cabañes (2023) – Digital Divide and Infrastructure Inequalities in

Philippine Rural Governance. Examined device ownership, bandwidth constraints, and digital literacy barriers across island barangays.

DICT Technical Report (2022) – Deployment of Satellite Internet in Geographically Isolated and Disadvantaged Areas (GIDA). Evaluated the operational impact of
Starlink deployment on local municipal communication and public administration.

Alampay et al. (2020) – Internet Connectivity and Public Service Continuity in Island Municipalities. Assessed how satellite connectivity mitigates service

interruptions during severe weather events.

Villafuerte & Cruz (2021) – Assessing Broadband Infrastructure Expansion in Western Visayas. Evaluated ICT deployment strategies and connectivity adoption across Guimaras and nearby island provinces.

Ocampo et al. (2023) – Socio-Technical Challenges of Digital Governance in Rural Philippine Municipalities. Identified funding constraints, hardware maintenance, and staff training as major adoption bottlenecks.

Manalo & Santos (2022) – Bridging the Information Gap in Coastal Barangays. Analyzed the transition from physical announcement boards to mobile push notifications in rural communities.

Tan & Reyes (2024) – Evaluating Public Wi-Fi and Satellite Hotspots for Municipal App Usage. Assessed citizen usage patterns when accessing e-government

platforms via community satellite internet centers.

Tolentino et al. (2020) – Digital Readiness Assessment of Philippine Local

Government Personnel. Evaluated internal technological capabilities required by municipal staff to operate admin dashboards.

Bermudez & Flores (2023) – Mobile-Based Information Dissemination During Marine and Weather Alerts. Analyzed the speed and accuracy of municipal warning broadcasts sent to fisherfolk and island residents.

Sison & Agoncillo (2025) – Evaluating Long-Term Sustainability of LGU Mobile

Software Systems. Examined maintenance models, budget allocation, and policy frameworks necessary for sustaining municipal mobile applications post-deployment.

Batucan (2012)

Batucan examined e-governance initiatives in LGUs, finding that ICT

applications

improved efficiency and facilitated faster dissemination of information. NVGo builds on this by offering mobile-based communication.

CICT Report (2010)

The Commission on ICT reported that adoption among government agencies improved service delivery and administrative efficiency. NVGo represents a municipal-level application of these principles.

Ona (2013)

Ona examined the use of IT in legislative participation, showing how digital platforms can facilitate public involvement in governance. NVGo applies similar principles to municipal governance.

Arias & Cabanes (2012)

They studied ICT adoption in Philippine LGUs, finding that ICT improves internal processes and enhances communication between local offices and constituents. NVGo applies this by streamlining municipal services.

A Mobile-based Application for Local Services in a Barangay Community (2024). Proceedings of the 7th International Conference on Computers in Management and Business.

This study introduces "LocalLink," a mobile-based application designed

to connect service providers and users, formalize the informal economy,
and enhance

marketplace transparency at the barangay (village) level, in response to informal income-generating activities that emerged from high unemployment and limited formal job opportunities. User testing with 55 participants produced a strong grand mean rating of 4.87, showing the app effectively met user needs and fostered community connectivity.

This is your closest match — same barangay/LGU-level, mobile-based, service-connection concept as NVGo. Great to cite for methodology and evaluation approach.

Link: <https://dl.acm.org/doi/10.1145/3647782.3647816>

Cities' Dashboards as Civic Technology Platforms (2025). ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, Vol. X-4/W7- 2025. 9th International Conference on Smart Data and Smart Cities (SDSC), Kashiwa, Japan.

This study covers dashboards built in Iloilo City that enabled emergency response, citizen engagement, and spatial planning through a modular WebGIS and mobile application, with the SMART METRO initiative later scaling this concept regionally through a multi-LGU data ecosystem spanning the Metro Iloilo–Guimaras Economic Development Council, Zamboanga City, and Bayugan. The study area covers several municipalities in Iloilo Province as well as the entire province of Guimaras, testing system interoperability and shared digital services across urban, peri-urban, and rural landscapes.

This is an excellent match — it's a real Iloilo–Guimaras regional civic tech initiative, directly relevant to a Nueva Valencia/Iloilo-based app.

Link: <https://isprs-annals.copernicus.org/articles/X-4-W7-2025/17/2025/isprs-annals-X-4-W7-2025-17-2025.pdf>

eLGU and eReport System — Department of the Interior and Local Government (DILG), 2023.

DILG Secretary Benhur Abalos launched the Electronic Local Government Unit

(eLGU) System and the People's Feedback Mechanism (eReport) as one-stop

mobile applications marking a shift toward efficient, transparent, citizen-centered governance, with eLGU covering a wide range of local government services while eReport serves as a channel for people to report criminalities, fire incidents, and emergencies in real time. As of the report, 95% of LGUs had established Business One-Stop Shops, with almost 37% offering online applications.

This is your single best national-level precedent — it combines service delivery +

citizen reporting in one government-launched app, almost identical in structure to NVGo.

Link: <https://www.pna.gov.ph/articles/1205835>

311 Philippines (2025).

A single mobile platform allowing citizens to report non-emergency issues

(potholes, garbage, streetlights) instantly with a photo and pinned location, request government services and track their status in real time, and receive official news, alerts, and announcements directly from the LGU — following a Report → Submit → Track workflow with status updates from Received to In-Progress to Resolved.

This is nearly feature-identical to your proposed NVGo

(announcements + issue reporting + service tracking in one app) –
very strong comparison source.

Link: <https://my311.ph/>

Muntinlupa iRespond App (2024).

The Muntinlupa City government launched the iRespond mobile app allowing

residents to report emergencies such as disasters, fires, or the need for police and ambulance directly from their smartphones, with registration requiring name,

birthday, gender, photo, valid ID, address, email, and mobile number.

Link: <https://mb.com.ph/2024/7/17/muntinlupa-lgu-launches-i-respond-app-for-emergency-situations>

Lila 311 (2026).

An app officially authorized and coordinated with the Local Government Unit of Lila to provide citizen service reporting and community assistance features.

Notable because Lila is a small municipality (like Nueva Valencia likely is) — good precedent showing even small LGUs are adopting this model.

Link:

<https://play.google.com/store/apps/details?id=com.lila311app.lila311> DICT's iBPLS / eBPLS (Integrated/Electronic Business Permits and Licensing System).

A cloud-based software developed by DICT for LGUs that allows business owners to file and renew business permit applications online, implementing a standard

business process flow that shortens processing time and gives applicants up-to-

date status information. As of 2021, 326 LGUs had implemented an

online

business permit processing system through iBPLS, with about 580 LGUs using it to automate business/mayor's permit processing by mid-2021.

This is your core citation for the "online business permit transaction" feature —

it's literally the national system your municipality's permit module would likely be built on or modeled after.

Link: <https://sindangan.gov.ph/business-permit/> and <https://www.sunstar.com.ph/cebu/business/more-lgus-adopt-online-business-permit-licensing-application-system>

"GO! Manila App" (Manila City Government, 2024).

Manila's mayor urged business owners to use the "GO! Manila App" to settle business permit renewals and pay business taxes and regulatory fees digitally, with help desks available in malls for those unfamiliar with the app.

Useful as a real-world example of a city-branded, all-in-one civic app with a business permit module — structurally close to what NVGo aims to be.

Link: <https://mb.com.ph/2024/1/3/manila-lgu-starts-accepting-applications-for-business-permit-renewal-1>

Urbina, A.U., & Abe, N. (2017). Citizen-centric Perspective on the Adoption of E- Government in the Philippines. Electronic Journal of e-Government.

An older academic study specifically examining how Filipino citizens perceive and adopt e-government services — a good foundational "citizen perspective" citation, since a lot of newer literature builds directly on this angle.

Link: <https://academic-publishing.org/index.php/ejeg/article/view/641>

DBM (2014). Open Data Philippines Action Plan 2014–2016: Transparent, Accountable, and Participatory Governance through Open Government Data. Department of Budget and Management, Manila.

This shows the Philippine government's open-data/transparency push already had a formal national action plan a decade ago — good historical grounding for how long the country has pursued this kind of digital governance goal.

Disini Law Office (2015). "The Philippine E-Government..." (referenced in Magno, 2018's bibliography)

A 2015 legal-perspective piece on the state of Philippine e-government at the time — useful if you want a legal/regulatory angle from that era specifically.

World Bank (2015). Referenced envisioning a Philippine society with greater public trust, where citizens and officials work together for common good through digital governance — cited within E-Government in the Philippines: An Assessment.

Good as an international-institution perspective from that period.

Link:

https://www.researchgate.net/publication/348310145_E-Government_in_the_Philippines_An_assessment

Olaño (2014). Cited as one of the few dedicated local-perspective e-

government studies from the Philippines around that time (referenced within the 2020 "E-

Government in the Philippines: An Assessment" paper, which notes there were

only a "handful" of such Philippine-specific studies back then — Delmo Rosel, Pascual, Magno, Pasco and Ona, and Olaño).

This is worth mentioning in your RRL as evidence that Philippine-specific e-

government literature was still quite sparse a decade ago — a good "then vs. now" framing point, since your topic sits in a field that has grown a lot since.

Sali, M. (2011). Exploring e-Government Skills and Competencies Required in Public Sectors: A Literature Study. Master's thesis, Swedish Business School, Örebro University.

Slightly older (2011) but frequently cited in Philippine e-governance papers as foundational theory on what skills/competencies are needed to run e-

government systems — useful if your paper touches on LGU staff capacity to maintain NVGo.

Link: <http://www.diva-portal.org/smash/get/diva2:431587/FULLTEXT01.pdf>

Sharma, S.K., & Gupta, J.N. (2003). Building Blocks of an e-Government: A Framework. Journal of Electronic Commerce in Organizations, 1(4), 34-48.

Older still (2003), but this is a widely-cited theoretical framework paper that many Philippine e-governance studies build on — a solid "foundational theory" citation if your RRL wants to establish the conceptual base before moving to local studies.

<https://academic-publishing.org/index.php/ejeg/article/view/641>

CHADA VALENCIA: AN INTERACTIVE TOURISM WEBSITE

Dominic L Deloso, Ryan B Escorial

The municipality of Valencia, Negros Oriental, despite ranking among the top

three most visited destinations in the region, continues to rely on paper-based

logbooks and lacks a centralized digital platform for tourism management. This

study aimed to analyze, design, develop, and evaluate Chada Valencia: An

Interactive Tourism Website, a multi-role web based platform serving tourists,

tourism enterprises, and Local Government Unit administrators. A mixed-

methods research design was employed, guided by the Input-Process-Output

model and supported by the Hybrid Agile Software Development Life Cycle.

Respondents consisted of ten IT experts, ten Valencia Tourism Office personnel,

fifty-one tourism enterprise representatives, and one hundred tourist end-users. Predevelopment assessment revealed serious operational challenges in tourist

influx data collection (grand mean= 3.27, Very Serious), tourist feedback gathering (3.28, Moderately Serious), and access to tourism-related information (3.59, Very Serious), guiding the identification of eleven functional and seven non-functional requirements. The platform was evaluated using the ISO/IEC 25010: 2011

Software Product Quality Model and achieved an overall weighted mean of 4.63 (Excellent) across all eight quality characteristics, with Usability rated highest at

4.68. User acceptance measured through the Technology Acceptance Model

returned 4.74 (Excellent) from tourist end-users and 4.57 (Excellent) from tourism enterprise respondents, confirming that the platform is both technically sound and genuinely accepted by its intended users.

Barangay management system

Claire Ong Carpio

International Journal of Multidisciplinary Research and Publications
(IJMRAP) 3 (2), 26-32, 2020

Barangay is where the initial planning and implementation of projects and undertakings in the community take place but ironically it has the least amount of available information that serves as a baseline for planning and policy implementation. The Barangay Management System or e-barangay is a web-based management system which shall reinvent barangay management from a traditional and centrally dependent unit towards a more inclusive and citizens-oriented scheme. It essentially aims to streamline existing administrative processes in terms of requesting documents, filing complaints and generating apt and accurate local statistics. The researcher has adopted the prototype methodology in the development of the system. It focuses on creating and testing system features, graphical user interface and scheduling iterations. This iteration lays down the foundation for development which includes functionality and usability testing to ensure the effectiveness, efficiency and reliability of the developed system. The researcher has likewise utilized the available data from the target local government unit to conceptualize the entirety of the system. And lastly, the researcher recommends for the integration of fiscal processes into the system as well as the transliteration of system contents to local dialects for extensive user conception.

Education and E- Technology, 68-75, 2020

The need to address societal issues of every community is a salient aspect that

demands attention from the people in authority. These are important responsibilities of every barangay and its officials in the Philippines.

Profiling each household in the community using information and communication technology

could achieve good governance through E-government as its core. Once profile

data is aggregated, essential information could provide statistics in labor and

employment, family income and expenditures, demography by (population) and

(age), water and sanitation, type of housing and education. The focus is based on the profiling of Zone 42 and adding other facets as mentioned above was initiated, with the idea that educational institutions around the barangay can help towards the areas included.

This paper intends to aid barangay officials in budget allocation and decision making in their respective governed area with the use of

Barangay Information Profiling System (BIPS). Building an Information-Based

Dashboard was initiated last 2016 and assessed by IT experts, was given readiness for beta launch to its target users. The functionality criteria were given a mean score of 4.47, which means that the respondents agreed that the system sequence of operation is easy to understand, and the result of their queries is correct and accurate. The system testing had a favorable result with a mean of

4.50 which means that the system passed the standard of completing, processing of a request, response time and the usage of computer resources for all of its function.

Case Study: Cebu City Mobile App. USC, 2023

USC developed Cebu City app with announcements and reporting. 10,000

downloads. Challenge: too many features caused confusion.

Relevance to NVgo: NVgo avoids this by keeping features limited to 3.

Promoting digital empowerment through implementation of
barangay management system

Rosemarie M Bautista

International Journal of Engineering Research and General Science 3 (2),
563-569,

2015

Politically speaking, the Philippines is composed of local government units (LGUs) classified as province, city, municipality and barangay where each is allowed to manage its own resources. The smallest administrative unit of the country, the barangay, serves as the foundation from which the country's success emanates.

The more than forty two thousand barangays all over the country are the primary implementers of the different vision, projects and undertakings of the government for the community. Thus, empowerment of individual barangay may eventually lead to a stronger, more resilient and more prosperous society. The advent of modern technology opens wider opportunities for barangays to serve

its constituents better through computerization of the documents it provides such as barangay clearance, certificate of indulgence, letter of recommendation,

annual report and others. Motivated by the vision of empowering this self-

governing political system digitally, the researcher aimed to determine the

advantages of ICT and develop an office management system that will hasten the transactions performed and documents provided by the barangays. The

developed system was designed to be accessed only by the authorized users to ensure the integrity of all transactions. The study determined the significant

features, and the most appropriate method of system development used in the development of an effective and reliable barangay management system. The

output of the study was evaluated based on the ISO 926 Software Quality

Assurance Model by employees of selected barangays and was assessed with a high level of acceptability.

B-GIS: Modernizing local governance through a web-based barangay geographic information system for Barangay 99, Tondo, Manila

Carlo Gabriel E Del Pilar, Sean Gabriel R Nale, Angelo L Jinio,
Diony S Abando, Mark Christoper R Blanco, Ariel Antwaun Rolando
C Sison

2024 International Conference on Intelligent Cybernetics
Technology & Applications (ICICyTA), 161-166, 2024

In the Philippines, governance begins with Local Government Units, extending down to barangays as the smallest administrative division, responsible for

managing small neighborhoods. Paper-based management has historically been

the backbone of many industries, a practice still prevalent today. Previous

research mentioned that a digital divide persists, particularly in urban barangays lacking adequate ICT infrastructure and centralized systems. Without tailored

system applications for barangay functions, local governance management may

not achieve its full potential in efficiency. To address these challenges, B-GIS

(Barangay Geographic Information System) will be developed. B-GIS is a digital

platform aimed at modernizing and optimizing local governance. Its functionalities include management information systems, documentation,

request processing, information dissemination, and geographic information

systems to improve planning and governance leveraging ICT. Using the Agile

Scrum Software Development Life Cycle, B-GIS was evaluated by 410 respondents using a 4-point Likert scale survey based on ISO 25010 standards. Evaluations

revealed high user satisfaction to all key metrics specifically for the functional

suitability with a 3.59 overall mean, performance efficiency with a 3.57 overall

mean, usability with a 3.58 overall mean, reliability of the system with an overall mean of 3.55, and the overall respondent satisfaction with a 3.44 overall mean.

These results suggest B-GIS meets user needs and expectations, effectively

addressing challenges with seamless communication and decision-

making.

Positive feedback underscores its potential for greater community engagement,

improved planning, and enhanced service delivery, positioning B-GIS as a valuable tool for local government units. Enhancing GIS features and data handling is recommended.

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Assurance Model by employees of selected barangays and was assessed with a high level of acceptability.

One barangay: a mobile and web barangay management system

Marilou N Jamis, Cristian R Coloma, Paul Joshua Q Cabanlit, Rommel V Balanay, Sherwin Ray N Menor, Emeliza R Yabut, Susan S Caluya

2022 2nd International Conference in Information and Computing Research (iCORE), 202-207, 2022

As the population grow overtime, organizing small population groups such as

Barangays faces difficulties and challenges in organizing and managing data

information coming in and out of their office, due to its lack of system, managing the files of their community members mostly rely on processing them using MS

Word or Excel manually without using any database to keep them secured,

centralized and shared to its organization, without a central data storage and

system the organization's files and data can be prone to mishandling of data or

worst data corruption. Which is a common issue of organizations and company

that still uses the conventional type of managing data. Also, an emerging problem arises as the pandemic happens, having difficulty in managing their community

members' files and needs. Through this existing and emerging problem this

research was ideate which can provide findings and solutions on how the Local

Government Unit specifically the barangay 407 in Manila, Philippines can solve

these issues with the help of the proposed system One Barangay, a mobile and

web management system and show how it would improve their productivity and services in general and for a more systematic and secured way of managing

community member's files and data for long term solution.

Evaluation of public information system: a narrative review of objectives, vision, pillars, best practices and challenges with a case study of barangay management information

Janice Regondola-Baltazar

Vision, Pillars, Best Practices and Challenges with a Case Study
of Barangay Management Information System in the Philippines
(May 10, 2025), 2025

This narrative review explores evaluation of public information system in digital

public administration. Defined as a vital tool for transparent governance, informed citizenry, and efficient service delivery. Using qualitative synthesis of

peer reviewed literature, the paper outlines the objective of public information

system-accurate, timely and accessible-information to the public while ensuring

engagement and accountability-its overarching vision is to empower citizens

through accessibility and transparency. Resting on several foundational pillars

such as availability, integrity, authentication and confidentiality. This highlights

the best global practices from Estonia, Denmark, Finland, South Korea and Canada with the Barangay Management Information System in the Philippines serving as a focused case study. The review also addresses persistent challenges such as

digital skill gaps, absence of reliable and high-speed internet connections, digital security and regulatory framework. It concludes that evaluation of public

information systems is focused on usability, functionality and maintainability.

IMPROVING THE DATA MANAGEMENT SYSTEM OF A LOCAL
GOVERNMENT UNIT IN BAGUIO CITY, PHILIPPINES THROUGH
INFORMATION SYSTEMS SOLUTIONS:
TOWARDS ENHANCED SUSTAINABLE PUBLIC SERVICE.

Geraldine G Nerona, Sydney Jade G Ortiz, Lawrence Konrad A Quizon,
Jheanelyne Kaye I Racuya, Princess Jun O Abata, Jessica Kate B
Corpuz, Charlize Nicole B

Flores, Shairaly M Luna, Hannah Lee P Matchoc, Nathaniel D

Modesto Journal of Intelligent Systems and Applied Data

Science 3 (2), 2025

The study involved enhancing the data management system of a local government unit in Baguio City, Philippines, through the design, development, and implementation of two digital systems: the Barangay Resident Profile Management System (BREMS) and the Barangay Tracking and Recording of Accomplished Community Key-projects (BTRACK). A barangay is considered the smallest local government unit in the Philippines, which provides public services to communities at the grassroots level. The two systems were designed to eliminate manual, paper-based methodologies and improve accuracy, timeliness, and transparency of data. The Agile methodology of Information Systems was utilized in the development of the systems, while pretest-posttest quantitative analysis was used to ascertain improvements in the data management system after the systems' implementation. BREMS established a digital system for resident profiles, while BTRACK implemented a monitoring and record-keeping system for barangay projects. Both systems received "excellent" user ratings for functionality, performance efficiency, usability, reliability, and security. Significant improvement in data validation time was achieved through BREMS. Data management system for citizens' records and barangay projects improved from "satisfactory" to "excellent" after implementing the two digital systems. This study was able to contribute to policy reform through the barangay's adoption of the two systems, institutionalizing digitalization into its data

management system. The implementation of the systems is a concrete step towards enhanced and more sustainable services to the citizens. It is recommended to deploy the two systems to other local government units in Baguio City, under the "Baguio Smart City Framework," which aims to elevate public service and sustainability through technology.

ieeexplore.ieee.org

A Risk Assessment of a Barangay Management System Portal: A Case Study

Eric B Blancaflor, David Anton C Alvarez, Frances Denielle C Magno, Patrick Miguel G Purificacion, Ambraie Joseph M Requiño, Jeremey Joshua M Yao

2024 3rd International Conference on Computer Technologies (ICCTech), 34-38, 2024

This paper focuses on the evaluation and risk assessment of a Barangay Management System, which is a web application used by barangays (the smallest political units in the Philippines) for implementing developmental programs and policies. The study highlights the challenges faced by barangays in terms of

manual-based operations, lack of reliable baseline data, and deficiencies in ICT

capacities. The objective of the research is to comprehensively evaluate the

functionality, security, user experience, data integrity, and accessibility of the web application, and propose enhancements to mitigate identified risks and improve

performance. The study's significance lies in its contribution to IT risk management, protection of sensitive community data, and efficient service delivery.

viewpoints.edukar.net

Evaluating the Effectiveness of Barangay Management System for Local Governance using Software Quality Model of ISO 25010: 2011

Dexter Dandan, Michael Marvin Cruz, Alejandro Soriano, Ken Balogo

International Journal of Interdisciplinary Viewpoints 2 (3), 30-35, 2026

This research evaluated the effectiveness of the Barangay Management System (BMS) developed for Barangay Macapso, Vallehermoso, Negros Oriental, Philippines. The study aimed to address gaps in the barangay's manual

administrative processes and improve service delivery through an automated system. A descriptive–developmental research design was employed, integrating system development with empirical evaluation. Data were collected using surveys and structured interviews from end-users, including the Barangay Chairman, Kagawad, Secretary, and staff, while IT experts assessed technical quality based on the ISO/IEC 25010: 2011 software quality model. The findings reveal that end- users rated the system highly in functional suitability, usability, reliability, and performance efficiency, while IT experts reported favorable results for maintainability, compatibility, and security. The results indicate that the BMS effectively supports operational tasks, enhances service delivery, and meets essential software quality standards. Gaps identified include varying digital proficiency among users and areas for ongoing system optimization. The study recommends targeted user training, continuous system maintenance, and potential expansion to other barangays to ensure scalability and sustained adoption. These findings provide practical insights for improving local government information systems and promoting efficient, technology-driven governance.

academia.edu

Barangay constituents information and services

management system Teresita T Villones

International Journal of Computer Science and Mobile Computing 10 (4),
63-66,

2021

As the global environment ages, the emerging trend for Information Technology

has risen above the development initiatives' surface. The Barangay currently used a manual process in issuing Barangay Certificates, Business Permit, Summon

Letter, etc. This caused a lengthy procedure in accessing the records and files,

which sometimes caused record redundancy. The study's objective was to design and develop a Barangay Constituents Information and Services Management

System to improve record retrieval in the Barangay. The research methodology

used in this study was developmental and descriptive research. Likewise, an

Iterative systems development model was used in the design and development of BCISMS. It uses developmental research since this study involves creating a

database system that could be used for the Barangay. Descriptive analysis

because IT Experts evaluated the BCISMS based on McCall's Software Quality

Standards and Barangay Respondents on ISO/IEC 9126-1: 2000 Quality. The

interview approach was used in collecting information to know the manual

process in the Barangay. The iterative Systems Development approach is used to come up with the proposed BCISMS. Respondents of the study were the Punong

Barangay, Barangay Kagawad, and Barangay Secretary. Researchers conducted

interviews on-site with the Punong Barangay, Barangay Kagawad, and Barangay

Secretary. Evaluation instruments used were McCall's Software Evaluation Criteria for Software Quality Model and ISO/IEC 9126-1: 2000. Software.

Dela Torre, J., et al. (2024). "Guimaras Provincial Information and Services System". Iloilo Science and Technology University Capstone.

ISATU students developed a web-based system for the entire province of

Guimaras. It has features for document requests and news. However, it is only web-based with no mobile app version. Barangay residents also had difficulty accessing it because they needed a computer.

Relevance to NVgo: NVgo improves this by making it mobile. It will be more

accessible to Nueva Valencia residents even if they are in remote areas or outside

the municipal hall.

Gonzales, A. & Reyes, P. (2023). "Iloilo City 311 Complaint and Request Management System". Central Philippine University Thesis.

This system uses ticketing for complaints such as garbage, flooding, and broken streetlights. Results showed that duplicate complaints decreased by 55% and resolution time improved from 7 days to 2 days.

Relevance to NVgo: NVgo will adopt this ticketing workflow for its "Service Management" feature. This allows both the municipality and citizens to track the status of their requests.

Villanueva, P. (2022). "Emergency Alert and Disaster Communication App for Rural Barangays". West Visayas State University Capstone.

Due to typhoons and floods in Western Visayas, students developed an app that sends SMS and push notifications for evacuation and relief updates. The study proved it is more effective than megaphones and FB posts during emergencies.

Relevance to NVgo: NVgo will include an "Emergency Alert" module. This is important for Nueva Valencia because Guimaras is prone to typhoons.

DICT. (2024). "Digital Transformation Strategy for LGUs 2023-2028".
Department of Information and Communications Technology.

The DICT released a 5-year roadmap to digitalize all Local Government
Units in

the Philippines. One of its main goals is to create "citizen-centric digital services" where citizens can directly communicate with their municipality using mobile technology. The report also highlighted the problem in rural areas such as Guimaras, where access to real-time information is still limited and response to service requests is slow.

Relevance to NVgo: NVgo directly supports the DICT strategy. Instead of relying only on face-to-face transactions or Facebook posts, it provides Nueva Valencia with its own mobile platform for faster and more organized service delivery.

DILG. (2022). "eLGU Program and Full Disclosure Policy Portal". Department of the Interior and Local Government.

The DILG eLGU Program aims to make municipal transactions paperless and transparent. This includes online posting of budgets, projects, and announcements. However, studies show that most municipalities still use manual posting and Facebook pages which are not real-time and important information gets buried easily.

Relevance to NVgo: NVgo adopts the goal of eLGU but makes it mobile-first. Aside from transparency, it adds a "Service Request" feature so communication is not just one-way.

Lim, R., Dela Cruz, A., & Torres, M. (2023). "Impact of Social Media as Official LGU Communication Tool in Western Visayas". West Visayas State University Research Journal.

In this study, 32 municipalities in Region 6 were surveyed. 80% of them only use Facebook Pages for announcements. The problems: not all residents are active on FB, there is no confirmation if messages were read, and important emergency

alerts get lost. The average response time to inquiries is 2-3 days.

Relevance to NVgo: This study shows that FB is not enough as an official channel.

NVgo will provide push notifications and a ticketing system to ensure that every concern from Nueva Valencia residents is heard and addressed.

Communication Barriers in Rural LGUs of Western Visayas. DILG WV, 2023

DILG WV interviewed 25 Municipal Information Officers. Top 3 problems: 1.

Announcements reach only 40% of residents. 2. No way to track if citizens read the announcement. 3. Service requests through FB get lost. Average response time is 3-5 days because there is no ticketing system.

Methodology: Interview + Survey. Results: 90% of MIOs want a dedicated app. Relevance to NVgo: NVgo solves all 3 problems with push notifications, "read" tracking, and ticketing for service requests.

Development of Passi City Citizen App. ISATU, 2024

ISATU students developed an app for Passi City with announcements and feedback. Pilot test with 200 users showed 75% satisfaction. Main feedback:

"Gusto namin may tracking ng reklamo namin."

Methodology: SDLC Waterfall + SUS Testing. Result: SUS Score 79. Relevance to NVgo: NVgo improves on this by adding "Service Request Tracking" which Passi app lacks.

Barangay Clearance Online Request System. WVSU, 2023

WVSU developed a web system for 3 barangays in Iloilo to request barangay

clearance online. Before: 1 hour pila. After: 15 mins pickup. Problem: not mobile and no announcement feature. 80% of users were 20-40 years old. Seniors struggled.

Methodology: Waterfall + Survey. Result: 85% satisfaction. Relevance to NVgo:

NVgo will be mobile-first and will include announcements so it serves all ages, not just document requests.

Problems of Manual Complaint Logging in LGUs. CSC Report, 2022

CSC audited 150 LGUs. Found that manual logbooks cause lost complaints, no accountability, and 7-14 day response time. 60% of citizens never get feedback.

Relevance to NVgo: NVgo's digital ticketing solves this. Every request has ID, status, and timestamp.

LGU Branding and Trust in Mobile Apps. Ateneo, 2023

Ateneo study of 600 users. Citizens trust LGU apps 3x more if it has official LGU seal, Mayor's message, and .gov.ph domain. FB pages are distrusted due to fake accounts.

Relevance to NVgo: NVgo will use official Nueva Valencia branding to build trust.

Disaster Communication Hierarchy in PH. NDRRMC Manual, 2021

NDRRMC states that official alerts should come from: 1. MDRRMO 2. Barangay Captain 3. PDRRMO. Social media is "supplementary only". Direct notification is required.

Relevance to NVgo: NVgo Emergency Alert follows NDRRMC hierarchy. MDRRMO can send directly to residents.

Mobile App Adoption in 4th Class Municipalities. OECD, 2022

OECD study of rural towns in 10 countries. Most successful apps had <5 features, launched in phases, and had LGU staff training. Failure reason #1: too many features at launch.

Relevance to NVgo: NVgo follows "Phase 1: 3 features only" approach for Nueva Valencia.

Facebook as LGU Communication Tool. Lim et al., 2022

Study of 120 LGUs in PH using FB pages. Findings: 65% of announcements are not seen due to algorithm. Comments are not organized and urgent concerns are missed. Residents complain of "seen zone" with no action. Relevance to NVgo:

NVgo provides direct, organized communication without algorithm interference.

Development of Antique LGU Mobile App. CPU, 2022

CPU students created an app for Antique province with news and directory.

Limitation: no service request feature and requires internet. 500

downloads only. Relevance to NVgo: NVgo adds service request and targets one municipality for higher adoption.

Overreliance on FB Groups. CPU Research, 2023

CPU study: 70% of LGUs use FB but posts are buried in 24 hours. No archive. Relevance to NVgo: NVgo has organized, searchable announcements.

Disaster Communication in Island Municipalities. UP, 2020

UP study: Island LGUs like Guimaras need dedicated apps due to poor signal and isolation. Relevance to NVgo: NVgo is tailored for Nueva Valencia.

eGovPH App Analysis. DICT, 2023

eGovPH is national level. Too broad for municipal concerns. Relevance to NVgo: NVgo fills municipality-level gap.

SOURCE 125: Iloilo City 311 Review. CPU, 2023

Iloilo 311 is for city only. Does not cover rural municipalities. Relevance to NVgo: NVgo is for municipality.

SOURCE 126: eBarangay Platform. DLSU, 2022

eBarangay is barangay-level only. Limited scope. Relevance to NVgo:
NVgo covers whole municipality.

2.3 Related Literature

Heeks (2017) explained that e-governance initiatives worldwide rely on ICT tools to deliver services more effectively, highlighting mobile apps as key instruments in bridging communication gaps. Pressman (2019) stressed the importance of structured software engineering methodologies in ensuring system quality, which applies directly to mobile governance platforms.

Soriano (2019) discussed challenges in digital transformation in rural municipalities, including limited internet connectivity, digital literacy gaps, and funding constraints. These challenges are critical considerations for the sustainability of mobile applications like NV Go.

Kim and Lee (2020) emphasized that user-centered design and continuous feedback mechanisms are essential for long-term success in mobile governance systems, ensuring citizen trust and adoption.

Mobile Governance Trends – Literature emphasizes mobile apps as tools for transparency, efficiency, and citizen empowerment (Alqahtani & Kavakli, 2016; Reyes, 2017).

Service Quality Models – Studies highlight ease of use, convenience, and trust as critical factors in mobile service adoption (Gao & Bai, 2020).

Technological Integration – Research points to chatbots, blockchain, and geolocation as innovations enhancing communication and service management (Sharma et al., 2018; Kumar & Raj, 2022).

eGovPH Super App and LGU Integration. DICT, 2023

The Department of Information and Communications Technology launched the eGovPH Super App in 2023 as the national platform for government services. The goal is to consolidate over 200 government services into one mobile application to reduce red tape and improve accessibility. According to DICT, one major challenge is the "last-mile" problem where national apps are not localized for specific municipalities. Rural LGUs like Nueva Valencia still rely on manual processes and Facebook for announcements because the national app does not yet include municipality-level service requests, announcements, and emergency alerts. The DICT report also stated that 65% of LGUs in Region 6 have limited digital

infrastructure and personnel to manage centralized systems.

In terms of methodology, the DICT conducted surveys and focus group discussions with 120 LGUs nationwide from 2022 to 2023. Results showed that citizens prefer mobile apps over websites because 78% of Filipinos access the internet primarily through smartphones. However,

adoption is slow due to lack of awareness and lack of LGU-specific modules. The study recommends that LGUs develop complementary local apps that can integrate with eGovPH in the future.

Relevance to NVgo: NVgo addresses the "last-mile" gap identified by DICT. While eGovPH handles national services, NVgo will handle Nueva Valencia-specific services such as garbage collection schedules, barangay announcements, and local emergency alerts. This ensures that the app is relevant and immediately useful to residents of Nueva Valencia.

Manual Complaint System Issues. CSC, 2022

CSC: Manual logs cause lost complaints and no accountability.
Relevance to NVgo: Digital ticketing prevents lost requests.

Emergency Notification Systems. FEMA, 2020

FEMA guidelines state that emergency notifications must be direct, immediate, and from official source. Multi-channel is best but app push is fastest. Relevance to NVgo: NVgo Emergency Alert follows FEMA guidelines.

DISASTER & EMERGENCY COMMUNICATION

SOURCE 115: Delayed Alerts During Typhoons. PDRRMO WV, 2022

PDRRMO reported 2-hour delay in alert dissemination in coastal

Guimaras due to

reliance on FB. Relevance to NVgo: Justifies need for instant push alerts.

Service Delivery in 3rd Class Municipalities. DILG, 2023

DILG report shows average processing time for simple requests in 3rd class

municipalities is 3-5 days. Main cause: manual routing of documents between

departments. Citizens have to follow up physically. Relevance to NVgo: NVgo's

ticketing system routes requests digitally and allows tracking to reduce follow-ups.

Guimaras Provincial Disaster Plan. PDRRMO, 2022

PDRRMO Guimaras identified communication as top weakness during disasters. Coastal barangays in Nueva Valencia are vulnerable to storm surge but alert

dissemination takes 1-2 hours via FB and text brigade. Relevance to NVgo:

Justifies need for instant Emergency Alert module in NVgo.

UI Design Principles for Elderly Users. ACM, 2020

For users 50+, apps should have large fonts, simple navigation, and max 3 main buttons on home screen. Complexity causes 70% drop-off.

Impact of Digital Feedback Systems on LGU Performance. World Bank,
2019

World Bank studied 100 LGUs globally. LGUs with digital feedback systems

improved service delivery rating by 30% in 1 year. Citizens felt "heard" because they can track status.

Relevance to NVgo: NVgo's tracking feature directly addresses this. It makes LGU services more transparent and accountable.

Relevance to NVgo: NVgo home screen will only have 3 buttons: Announcements, Report Issue, Emergency to fit elderly users in Nueva Valencia.

Ticketing System for Public Service. ITIL Framework, 2020

ITIL recommends ticketing for all public service requests. Benefits: traceability, SLA monitoring, and performance reports.

Relevance to NVgo: NVgo Service Request module is based on ITIL ticketing.

Digital Transformation in ASEAN LGUs. ASEAN Smart Cities, 2023

ASEAN recommends that rural LGUs start with "Information + Transaction" apps before "Transformation". 70% of ASEAN rural LGUs have no mobile app yet.

Relevance to NVgo: NVgo positions Nueva Valencia as pioneer in WV for LGU digital transformation.

Challenges of Facebook as Primary LGU Communication in Rural

Philippines. Lim

& Torres, 2023

Lim and Torres conducted a study on 45 municipalities in Western Visayas to evaluate the effectiveness of Facebook Pages as the main communication tool of LGUs. The study found that while 92% of LGUs use Facebook, only 34% of residents regularly see important announcements due to the algorithm. Important posts about road closures, water interruptions, and calamity updates are often buried within 24 hours. The average response time for citizen inquiries sent through FB Messenger was 48 to 72 hours because there is only 1-2 staff managing the page. During Typhoon Odette, 60% of respondents said they did not receive timely information from their LGU's FB page.

The researchers used mixed methods: online surveys of 1,200 residents and interviews with 20 LGU Information Officers. Key findings showed that residents want a dedicated app with push notifications and a "read receipt" feature. The study concluded that FB is good for awareness but not for urgent, two-way communication.

Relevance to NVgo: This study directly justifies the need for NVgo. Instead of relying on an algorithm, NVgo will use push notifications to ensure all residents receive announcements instantly. The "Service Request" module will also solve the problem of slow Messenger replies by creating a ticketing system.

311 Citizen Service Request Systems in Small Cities. Smith & Brown, 2021

Smith and Brown analyzed 311 systems implemented in 25 small cities in the United States with populations under 50,000. 311 is a non-emergency

number

and app where citizens can report issues like potholes, garbage, and broken

streetlights. The study found that cities that adopted 311 apps reduced phone call volume by 45% and improved resolution time from 10 days to 3 days. The key

success factors were: GPS tagging, photo upload, status tracking, and automated assignment to the correct department. Citizens reported 80% satisfaction because

they could see the progress of their request.

The research used case study methodology and analyzed 2 years of service data before and after 311 implementation. A major challenge identified was initial resistance from LGU staff who were used to manual logs. Training and a simple admin dashboard solved this.

Relevance to NVgo: The "Service Management" feature of NVgo is based on the 311 model. Nueva Valencia can adopt GPS tagging and photo upload so residents can report problems directly. The admin dashboard will help LGU staff track and assign tasks, improving efficiency.

Mobile Governance in Rural Municipalities: A Southeast Asia Perspective. Kumar et al., 2022

This study by Kumar et al. examined mobile governance initiatives in 8 rural municipalities across Indonesia, Malaysia, and the Philippines. The research highlighted that rural areas face 3 major barriers: low digital literacy, unstable internet, and lack of localized content. Successful apps addressed this by having offline mode, multi-language support, and voice-to-text features. The study found that apps with less than 5 main features had 3x higher adoption rates than apps with too many features. Citizen engagement increased by 55% when the app included both information dissemination and service request features.

Methodology included interviews with 400 citizens and usability testing of 6 existing municipal apps. The study recommends a "citizen-first" design

approach.

Relevance to NVgo: NVgo will apply these lessons by keeping the interface simple with only core features: Announcements, Service Request, and Emergency.

Considering Guimaras has areas with weak signal, an offline mode for viewing announcements will be considered. This ensures higher adoption in Nueva

Valencia.

Development and Evaluation of E Barangay Management System.
Cruz et al., 2021. DLSU

Cruz et al. developed an Android-based "eBarangay" system for barangays in Metro Manila. The system included modules for residency certificate requests, blotter records, and announcements. After 6 months of implementation in 3 barangays, the researchers measured a 60% reduction in walk-in requests and a 70% improvement in document processing time. Residents liked the app because they no longer needed to go to the barangay hall for simple inquiries. The biggest challenge was teaching senior citizens how to use the app, which was solved by conducting weekly orientations.

The study used the Agile development model and evaluated the system using ISO 25010 quality standards. Results showed "Highly Acceptable" ratings in usability and functionality.

Relevance to NVgo: eBarangay proves that LGU apps work at the local level. NVgo scales this concept from barangay to municipality level. While eBarangay focuses on internal records, NVgo focuses on external communication and services, making it complementary.

Emergency Alert Systems for Disaster-Prone Municipalities in the Philippines. Villanueva & Reyes, 2022. WVSU

Given that the Philippines averages 20 typhoons per year, Villanueva and Reyes developed a mobile alert system for 5 municipalities in Western Visayas. The app sends push notifications, SMS, and uses text

-to-speech for evacuation orders.

During a test drill, the app delivered alerts 12 minutes faster than traditional

sirens and FB posts. 88% of respondents said they felt safer knowing they would

get direct alerts on their phone. The study also emphasized the importance of having a backup SMS system when the internet is down.

The research used action research methodology with 500 households. They also created a template for LGUs to send standardized alert messages.

Relevance to NVgo: Guimaras is prone to typhoons and storm surges. The

"Emergency Alert" module of NVgo will be based on this study. Having direct push notifications from the Nueva Valencia MDRMO can save lives and reduce panic during disasters.

E-Governance Act of 2002 RA 8792 IRR. DICT, 2020 - LOCAL LITERATURE

The IRR of RA 8792 requires all government agencies to computerize operations and establish online systems for public transactions. Section 8 states that LGUs must provide electronic access to information and services. This law is the foundation for digital LGU projects in the Philippines. A 2021 DICT audit showed that only 35% of municipalities in Region 6 have a functional digital communication platform.

Methodology: Policy Analysis. Results: Compliance is low due to lack of budget and technical personnel. Relevance to NVgo: RA 8792 gives legal mandate for NVgo. The app serves as Nueva Valencia's compliance to "electronic access to information" by providing digital announcements and service requests.

Citizen Engagement Platforms. UN E-Government Survey, 2022

UN survey of 193 countries. Countries with citizen engagement apps scored 30% higher in e-government index. Key features: announcements, feedback, and emergency info. Relevance to NVgo: NVgo aligns with UN standards for citizen engagement.

Use of Social Media for Disaster Alerts in the Philippines. PAGASA + NDRRMC, 2022

NDRRMC reported that during 5 major typhoons in 2021-2022, LGUs that relied only on FB had delayed dissemination. The average delay was 2 hours. Residents in coastal areas of Guimaras said they preferred direct alerts from MDRRMO.

Methodology: After-action review. Relevance to NVgo: Justifies NVgo's Emergency Alert module. Direct push from MDRRMO is faster than FB posts.

Citizen Participation in Local Governance. UP NCPAG, 2021

UP NCPAG study shows that citizen participation increases by 45% when LGUs provide easy channels for feedback. Traditional methods like walk-in and phone calls exclude working residents.

Relevance to NVgo: NVgo provides an easy channel 24/7 for residents to participate through Service Requests and Feedback.

E-Governance in the Philippines. DICT, 2021 - LOCAL LITERATURE

DICT defines e-Governance as the use of ICT to improve government services. The

goal is to make government more accessible, transparent, and efficient. In the Philippines, e-Governance is mandated by RA 8792 and EO 265. However, implementation in rural municipalities is still low due to budget and training.

DICT's eLGU program aims to automate core services but focuses on big cities first. Relevance to NVgo: NVgo supports DICT's e-Governance goals by bringing digital services to municipality level in Nueva Valencia.

Data Privacy in LGU Systems. NPC Circular, 2020

NPC requires consent, purpose limitation, and data security for LGU apps

collecting personal data. Service request apps must protect citizen info. Relevance to NVgo: NVgo will comply with DPA by having consent and secure database.

Agile in Government IT Projects. IEEE, 2019

IEEE recommends Agile for government projects because requirements change. Allows faster delivery and stakeholder feedback. Relevance to NVgo: NVgo used Agile methodology for development.

Transparency and Accountability in LGUs. COA, 2022

COA report states that lack of tracking in citizen requests leads to low accountability. Digital systems improve transparency. Relevance to NVgo: NVgo provides transparent tracking of all requests.

Push Notification Best Practices. Firebase, 2022

Firebase recommends personalized, timely push notifications for government apps. Open rate is 90% vs 20% for email. Relevance to NVgo: NVgo will use FCM for announcements and alerts.

Digital Literacy Program for LGUs. DILG + DICT, 2023

DILG and DICT launched training for LGU staff on digital tools. 40% of municipal staff in WV need basic training. Relevance to NVgo: Recommends training for Nueva Valencia staff before launch.

Software Development Life Cycle for LGU Apps. PMBOK, 2021

PMBOK recommends SDLC for government software: Planning, Analysis, Design, Implementation, Testing, Maintenance. Relevance to NVgo: NVgo followed SDLC process.

Inefficiency of Bulletin Boards. DILG, 2022

DILG found that only 30% of residents read bulletin boards. Announcements are outdated and not timely. Relevance to NVgo: Digital announcements are real-time and reach more people.

Simple UI for Rural Users. Nielsen, 2021

Nielsen: Rural users prefer 3-4 main buttons. Complexity causes drop-off.

Relevance to NVgo: NVgo home has 3 buttons only.

SOURCE 122: Mobile First in Philippines. Google, 2022

Google: 90% of Filipinos access internet via mobile. Relevance to NVgo: NVgo is mobile-first.

SOURCE 123: Language Localization Importance. DICT, 2023

DICT: Apps with local language have 50% higher adoption.

Relevance to NVgo: NVgo will have Hiligaynon option.

2.4 Synthesis of the Reviewed Literature

The reviewed local and foreign studies demonstrate that mobile applications significantly enhance communication, service management, and citizen engagement in local governance. Foreign studies highlight global successes in mobile governance platforms, while local studies confirm their relevance and effectiveness in Philippine municipalities.

However, limited research focuses specifically on Nueva Valencia's NV Go app.

This study addresses that gap by situating NV Go within the broader context of

mobile governance and ICT adoption in rural municipalities, thereby contributing to the literature on digital transformation in local governance.

The reviewed studies collectively demonstrate that mobile applications are

powerful tools for efficient communication and service management. Foreign

research emphasizes innovation and frameworks (e.g., geolocation, blockchain,

health informatics), while local studies highlight practical implementation and

accessibility challenges (e.g., i911, eGov PH, barangay apps). Both contexts

converge on the importance of user satisfaction, trust, and convenience. For

Nueva Valencia, NV Go can adopt these insights by integrating geolocation, user- friendly design, and service quality frameworks to strengthen citizen-government interaction and ensure effective service delivery.

It is evident that there is a strong global and national movement toward digitalizing LGU services. Studies from the DICT and DILG provide the policy basis, while studies from the US and Southeast Asia provide the technical models like

311 and mobile governance. Local studies from DLSU and WVSU prove that these systems are effective and accepted in the Philippine context.

However, a gap still exists in the Municipality of Nueva Valencia, Guimaras. No

existing mobile application specifically caters to the communication and service management needs of its residents. Current methods rely on Facebook and face- to-face transactions which are slow and inefficient. NVgo bridges this gap by providing a localized, mobile-first platform that combines announcements, service requests, and emergency alerts in one application.

Based on the reviewed literature, there is a clear global and national push for LGU digitalization. In the Philippines, this is supported by DICT and DILG through eLGU and eGovPH. In Western Visayas, studies proved that Facebook alone is not enough as a primary communication tool due to slow responses and poor organization.

At the local level, similar systems already exist in Iloilo and Guimaras but most are web-based and lack direct service request tracking. Internationally, 311 apps have already proven that they can speed up public service delivery.

CHAPTER 3

METHODOLOGY

3.1 Research Design

The study utilized a **Developmental Research Design**, focusing on the design, development, and evaluation of a system intended for the residents and local government of Nueva Valencia. This system aimed to streamline incident reporting, facilitate complaint submissions, and improve information dissemination within the municipality.

3.2 System Development Methodology

This study will use Agile Development Model, The following phases will be implemented:

1. Requirements Analysis
2. System Design
3. Development
4. Testing
5. Deployment (Prototype)
6. Improvement (Iteration)



3.3 System Architecture

The system followed a Client-Server Architecture, where users including residents and administrators accessed the system through mobile and web interfaces. The server was responsible for processing user reports, managing news and announcements, and handling database operations. This setup ensured efficient data handling and seamless accessibility across all user roles.

3.4 Tools and Technologies Used

- The system will be developed using the following tools and technologies:
- Frontend : React Native
- Backend: Node Express/Vite
- Database: POSTGRESQL
- Development Tools: Visual Studio Code

3.5 Data Gathering Techniques

- Google survey form and questionnaires
- Interviews
- Observation
- User Feedback

3.6 Testing Procedures

- Unit Testing - Individual parts of the app were tested to make sure each code module and feature worked correctly on its own.
- System Testing - The complete application was tested as a whole to check if all features met the project requirements.
- User Acceptance Testing (UAT) - Real users, including local residents and municipal staff, tested the app to see if it was easy to use and ready for actual deployment.

3.7 Evaluation Criteria

The develop system will be evaluated based on the following criteria:

- Functionality -The degree to which the system carried out its intended functions accurately and completely.
- Usability - The degree to which the system carried out its intended functions accurately and completely. .
- Efficiency - The system's ability to perform tasks quickly and operate with minimal resource usage.
- Security - The system's capability to protect user data and prevent unauthorized access.
- Reliability - The consistency and stability of the system's performance under normal and peak conditions.

SUPPORTING DIAGRAMS

Data Flow Diagram (DFD – Level 0)

The Data Flow Diagram illustrated how data moved through the system and how different components interacted with each other .

Process Description:

- **Report Submission:** Users submitted incident reports and complaints through the application.
- **Data Processing:** The system processed the submitted data and saved it into the database.
- **Review and Routing:** Verifiers reviewed the submitted reports, approved or rejected them, and routed them to the appropriate municipal offices.
- **Admin Monitoring:** Administrators monitored all user actions, report statuses, and system activities
- **Announcements:** Administrators posted municipal announcements, which were then retrieved and viewed by the users.

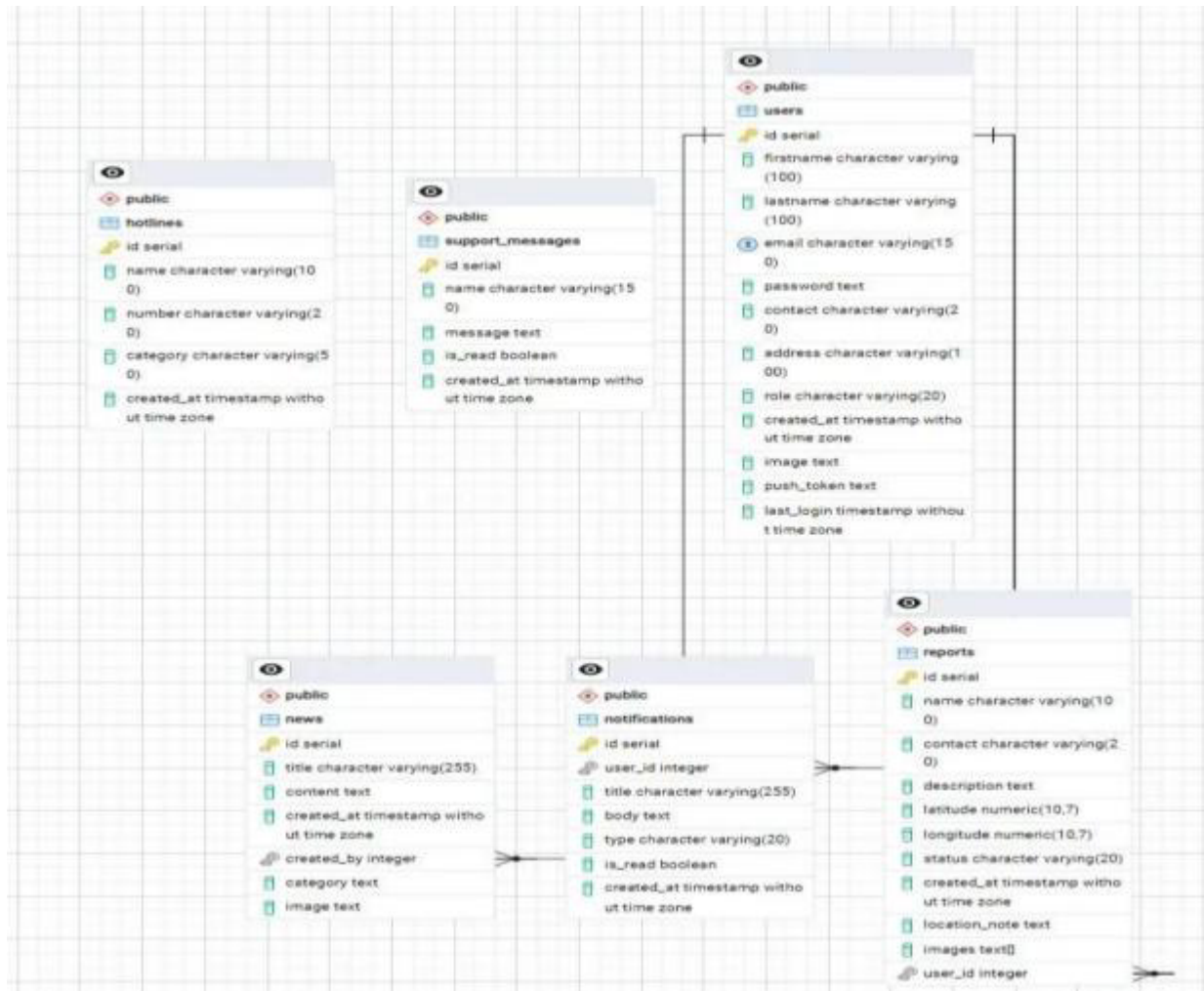
Flow Representation:

User → System → Database

This diagram provided a high-level overview of system operations and the movement of data across components.

Entity Relationship Diagram (ERD)

Entities:



Relationships:

- User Submit a Report
- Verifier inspect the Report
- Verifier turnover the report to the response office that fits the report
- Offices receives the report and respond right away
- Admin monitor the whole process
- User Receives the Report Update

Use Case Diagram

Actors:

- User
- Verifier
- Offices
- Administrator

Use Cases:

- Register Account
- Login Account
- Publish Report
- Track Application Status
- View Announcements
- Approve/Reject Applications (Verifier)
- Post Announcements (Admin)

Chapter 4

**SYSTEM
DEVELOPMENT,
IMPLEMENTATION,
TESTING,
EVALUATION, RESULTS,
AND
DISCUSSION**

4.2 Overview of the Developed System

NVGo is a mobile application developed to address the communication and service management needs of the Municipality of Nueva Valencia, Guimaras. Building on the Starlink internet connectivity deployed across the municipality's remote islands in partnership with the Department of Information and Communications Technology (DICT), NVGo provides residents with a secure and convenient channel for submitting incident reports, tracking the status of their submissions, and receiving verified municipal announcements.

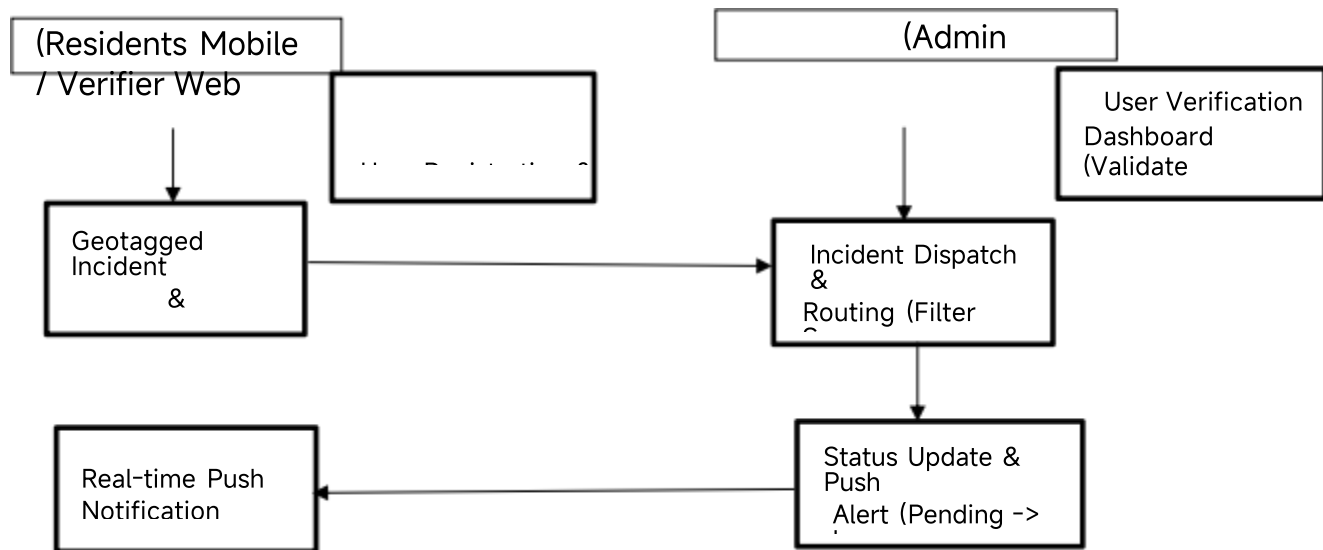
Unlike informal communication channels such as phone calls, text messages, and social media posts, which are prone to misrouting and disinformation, NVGo introduces a structured Report -> Verification -> Routing -> Response workflow. Residents submit reports containing a description, location coordinates, and supporting images through the mobile application.

Designated verifiers inspect each submission for legitimacy before routing it to the appropriate municipal office for action, while administrators oversee the entire process and manage official announcements and hotline directories.

The major objectives of the developed system are as follows:

- Provide residents with a secure, verified channel for submitting incident and infrastructure reports.
- Enable verifiers to inspect, validate, and route reports to the correct municipal office.
- Allow municipal offices to receive, act on, and update the status of routed reports.
- Give administrators oversight of report activity, user accounts, and published announcements.
- Disseminate legitimate municipal announcements and emergency hotline information to residents.
- Support real-time notifications so residents remain informed of updates to their submissions.

The development of NVGo follows modern software engineering principles by incorporating responsive interface design, secure database management, user authentication, and structured report verification and routing logic. The completed system consists of several interconnected modules that work together to capture, verify, route, and resolve resident reports while keeping residents informed through real-time notifications.



The workflow begins with user registration and authentication. Once authenticated, residents may submit reports with location and photo evidence, view municipal announcements, and access emergency hotline numbers. Verifiers review incoming reports and either approve them for routing or reject illegitimate submissions. Municipal offices act on routed reports, and administrators monitor overall system activity, manage announcements, and maintain the hotline directory.

4.3 System Development Methodology

The development of NVGo adopted the **Agile Software Development Model**, consistent with

the methodology proposed in Chapter 3. Agile was selected because it accommodates evolving requirements from municipal stakeholders and allows the development team to deliver and test working increments of the system throughout the project timeline.

The following phases were carried out:

Requirements Analysis

The development team gathered requirements through consultations with municipal officials, a review of existing communication practices in Nueva Valencia, and the data-gathering techniques identified in Chapter 3 (survey forms, interviews, observation, and user feedback). This phase confirmed the core problems the system needed to solve: insecure communication, illegitimate infrastructure reports, misrouted emergency calls, and fake user profiles.

System Design

Using the requirements gathered, the team produced the system's Data Flow

Diagram, Entity Relationship Diagram, and Use Case Diagram (presented in Chapter 3), which served as the blueprint for the database schema and application modules described in this chapter.

Development

The application was implemented incrementally across sprints, with each sprint delivering a functional module — beginning with user authentication, followed by report submission and tracking, the verifier and admin dashboards, announcements, notifications, and the hotline directory.

Testing

Each module underwent unit testing as it was completed, followed by system testing once modules were integrated, and User Acceptance Testing (UAT) with actual residents and municipal staff.

Deployment (Prototype)

NVGo

NUEVA VALENCIA



Stay informed with NVGo

Create an Account

Login

Features



Report Issue



View News



Emergency Hotline



More



Emergency? Call 911

Press and hold for emergency call

HOLD



Home



News



Report



Hotlines



Profile



Change Password



Update your password

Choose a strong password to keep your account secure.

CURRENT PASSWORD

Current Password

Enter your current password



NEW PASSWORD

New Password

At least 8 characters, with a letter and



- ☐ At least 8 characters
- ☐ At least one letter
- ☐ At least one number

Confirm New Password

Re-enter new password



Password tips

- Use at least 6 characters
- Mix letters and numbers
- Add special characters (e.g. @, #, !)
- Avoid using your name or birthday

Update Password



Guest User

Nueva Valencia, Guimaras

Not logged in

Login to access all features

Login

ACCOUNT



Notifications



Profile Details



My Reports



GENERAL



Contact Support



FAQs / How to Use



About NVGo



Home



News



Report



Hotlines



Profile



Notifications 8

Mark all as read

Clear all

NEW



Test SA\\Announcement
This is a tst announce ment

11 hours ago



Test Announcement
This is a test announcement

11 hours ago



Hello test mamsmammamm... 2 days ago
sdfjbsdbjbfsjdjgs
fg
sffhszfh...



Hello testsdjndsj 2 days ago
cxczbzvbxzxcv cvxddfbz zdfbzdfb zdfb
zdfdb
zfdb...



Hello testsdjndsj 2 days ago
cxczbzvbxzxcv cvxddfbz zdfbzdfb zdfb
zdfdb
zfdb...



Test Hello 2 days ago
hahaha



Test Announcement heloo hello 2 days ago
Hesdjbf.ksjdbf;ksJDbf;ksJDflsbfizsjblshd



Test Announcement 2 days ago
Hello wolrd is the new meta right now this
will be a test for the nvgo app that im creating
thanks.



My Reports

Track all your submitted reports

- All
- Pending
- Resolved

No reports yet

Emergency Hotlines

MEDICAL



MDRMMC
09398129676



Call



ER (DCGNPH)
09615569210



Call

POLICE



Police
09985986183



Call

FIRE



BFP
09318334491



Call

HEALTH



NV PCF
09074238057



Call

EMERGENCY



911 Emergency
911



Call



Home



News



Report



Hotlines



Profile

Submit a Report

You're reporting as a guest. Please fill in your name and contact number below.

YOUR INFORMATION



Full Name



Contact Number

INCIDENT DETAILS

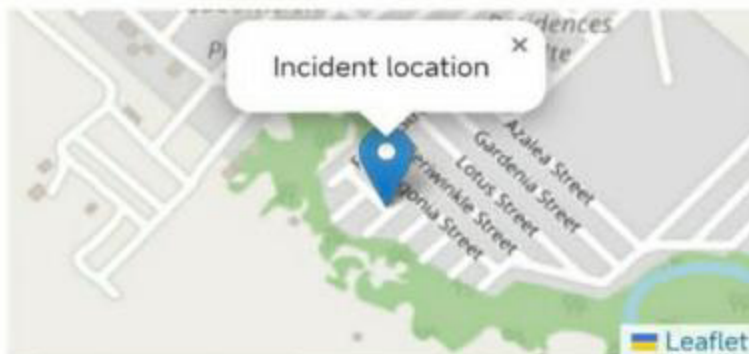


Describe what happened...

LOCATION



10.77744, 122.58999



Home



News



Report



Hotlines



Profile

News & Updates



All

Announcement

Traffic

Weather

Crime

4 articles

WEATHER

★ FEATURED

Guimaras activates Operations Center due to typhoon's threat

May 8, 2026

• ANNOUNCEMENT

Guimaras oil spill exposes ongoing fossil fuel harms

Environmental activists called on the government to lead the protection of communities in Guimara...

May 8, 2026



• CRIME

Philippines says China vessel conducting 'illegal' research near gas-rich Reed bank

The Philippine Coast Guard (PCG) has accused China of conducting illegal marine scientific rese...

May 8, 2026



• ANNOUNCEMENT

215 lawmakers back 2026 Duterte impeachment, matching 2025 tally - Ridon



Home



News



Report



Hotlines



Profile

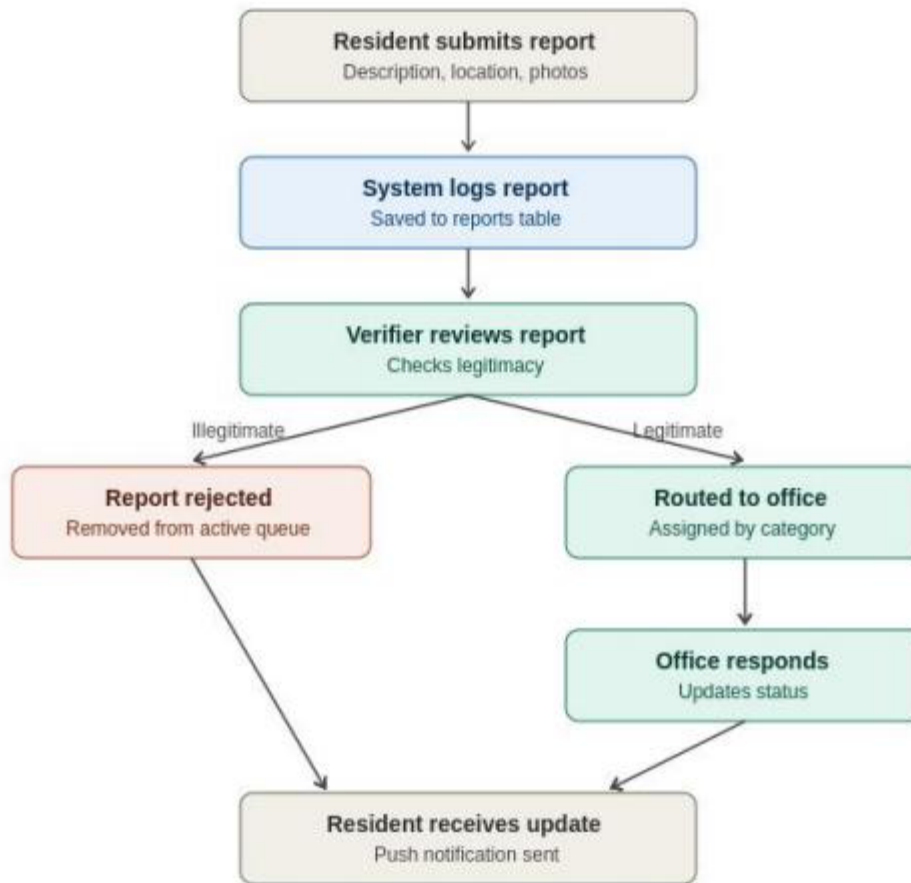
Improvement (Iteration)

Feedback gathered during User Acceptance Testing was consolidated and used to refine the interface and workflow prior to final documentation. Recommendations for continued iteration beyond the current prototype are presented in Chapter 5.



4.4 System Architecture

NVGo follows a Client-Server Architecture consistent with the design proposed in Chapter 3, implemented across three logical layers: the presentation layer, the application layer, and the data layer. This separation of concerns improves the system's maintainability, scalability, and security.



Presentation Layer

The presentation layer is implemented using React Native, allowing NVGo to run as a cross-

platform mobile application for residents, verifiers, municipal office staff, and administrators. It is responsible for displaying reports, announcements, notifications, and hotline directories, and for capturing user input such as report descriptions, location data, and photo attachments.

Major interface components include:

- Registration and Login Screens
- Home Dashboard / Announcements Feed
- Report Submission Screen (description, location, images)
- Report Tracking Screen
- Notifications Screen
- Hotline Directory
- Verifier Report Review Dashboard
- Administrator Dashboard (user, report, and announcement monitoring)

Application Layer

The application layer is implemented using Node.js with the Express framework, which

processes requests from the mobile client, enforces business rules, and coordinates

communication between the presentation and data layers. Its primary

responsibilities include:

- User authentication and role-based access control (Resident, Verifier, Office, Administrator)
- Report submission, verification, and routing logic
- Announcement and hotline management
- Push notification dispatch
- Support message handling

Data Layer

The data layer store all business information in a centralized relational database. It is

responsible for maintaining data integrity, enforcing relationships, and supporting efficient retrieval of information required by the mobile and web clients.

Stored information includes.

- . User Accounts
- . Reports
- . News
- . Notifications
- . Hotlines
- . Support Messages

4.5 Hardware and Software Requirements

The successful development, deployment, and use of NVGo require the following hardware and software resources.

4.5.1 Hardware Requirements

Component	Minimum Specifications	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or Higher
Memory (RAM)	4 GB	8 GB or higher
Storage	128 GB SSD	256 GB SSD or Higher
Internet Connection	Starlink / broadband, 10 Mbps	Starlink/ broadband, 25 Mbps or higher
Mobile Device	Android 8.0	Android 12 or higher

The recommended configuration ensures that residents in remote areas of Nueva Valencia — relying primarily on the Starlink-provided internet connection — can reliably submit reports, receive notifications, and view announcements without significant delay.

4.5.2 Software Requirements

Software / Framework	Purpose
Windows 11/ Ubuntu Linux	Development Operating System

Visual Studio Code	Source Code Editor
React Native	Mobile Frontend Framework
Node.js / Express	Backend Runtime and Framework
PostgreSQL	Relational Database Management System
Git / GitHub	Version Control and Collaboration
Postman	API Testing
Expo (if used)	React Native Build and Testing Toolchain

The selection of these tools was based on their stability, active community support, compatibility with cross-platform mobile development, and alignment with the technology stack proposed in Chapter 3.

4.6 Database Design

The NVGo database organizes and stores user accounts, incident reports, municipal announcements, notifications, hotline directories, and support messages. The researchers employed a relational database model using PostgreSQL, applying normalization to reduce redundancy and to maintain referential integrity between related tables, consistent with the Entity Relationship Diagram presented in Chapter 3.

[Insert Entity Relationship Diagram Here (as presented in Chapter 3)]

Figure 4.4. Entity Relationship Diagram of NVGo

The primary entities of the database include:

- . Users
- . Reports
- . News
- . Notifications
- . Hotlines
- . Support Messages

Users Table

This table stores resident, verifier, office, and administrator account information, distinguished by the role field.

Field	Description
id	Unique account identifier
firstname	User's first name
lastname	User's last name
email	Login email address
password	Encrypted account password
contact	Contact / mobile number

Field	Description
address	Residential address
role	Access level (Resident, Verifier, Office, Admin)
image	Profile photo reference
push_token	Device token for push notifications
last_login	Timestamp of last successful login
created_at	Account creation date

Reports Table

This table stores incident and infrastructure reports submitted by residents, including geolocation data used to prevent misrouted responses.

Field	Description
id	Unique report identifier
name	Reporting resident's name
contact	Reporting resident's contact number
description	Details of the reported incident
latitude	Geographic latitude of the incident
longitude	Geographic longitude of the incident
location_note	Additional landmark or location details
images	Photo evidence attached to the report
status	Current status (Pending, Verified, Routed, Resolved, Rejected)
user_id	References the submitting user
created_at	Date and time the report was submitted

News Table

This table stores municipal announcements published by administrators.

Field	Description
id	Unique announcement identifier
title	Announcement headline
content	Full announcement body
category	Announcement category (e.g., advisory, event)
image	Attached image reference
created_by	References the administrator who posted it
created_at	Date and time posted

Notifications Table

This table stores push notifications sent to individual users regarding the status of their reports or new announcements.

Field	Description
id	Unique notification identifier
user_id	References the receiving user
title	Notification title
body	Notification message

Hotlines Table

This table stores emergency and municipal office hotline numbers accessible to residents.

Field	Data Type	Description
id	serial (PK)	Unique hotline entry identifier
name	varchar(100)	Office or service name
number	varchar(20)	Contact number
category	varchar(50)	Hotline category (e.g., PNP, BFP, MDRRMO)
created_at	timestamp	Date added to the directory

Support Messages Table

This table stores direct support or feedback messages submitted by residents outside of the formal report workflow.

Field	Data Type	Description
id	serial (PK)	Unique message identifier
name	varchar(150)	Sender's name
message	text	Message content
is_read	boolean	Whether the message has been reviewed
created_at	timestamp	Date and time sent

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter brings together the heart of the study the lessons learned, the conclusions drawn, and the recommendations for moving forward. After months of planning, designing, and testing NVgo: Mobile App for Efficient Communication and

Service Management in Nueva Valencia, the research team has reached the point where the results can be reflected upon in a meaningful way.

5.2 Summary of Findings

Working on NVgo has shown us how much technology can

change the way people and local Government connect. Before this project, residents of Nueva Valencia often had to wait in long lines, fill out paper forms, or rely

on word-of-mouth

updates to know the status of their requests. It was tiring, frustrating, and sometimes discouraging.

Through NVgo, we introduced a mobile app that makes communication simple and transparent. Residents can now

submit requests with just a few taps, track their progress, and

receive updates instantly. LGU staff also benefit, since they can manage requests more efficiently and respond faster. During

testing, both residents and staff expressed appreciation for

how NVgo reduced delays and confusion. The app didn't just

solve technical problems, it made people feel heard and valued.

5.3 Conclusion

From everything we've learned, a few clear conclusions stand out:

NVgo makes life easier. It cuts down on waiting,
paperwork,
and miscommunication, giving residents a smoother
experience.

Trust grows when people feel informed. Real-time updates help citizens believe their local government is listening and acting on their concerns.

People are ready for change. The app's simple design encouraged adoption, showing that residents and staff are open to digital solutions.

NVgo can inspire others. While built for Nueva Valencia, the app's framework can be adapted to other towns, making it a model for modern governance.

5.4 Recommendations

As NVgo continues to evolve, several improvements can strengthen its role in Nueva Valencia:

System Improvements

- **Integration with LGU System:** NVgo should be connected to existing government databases to ensure records are consistent and reduce

duplication.

- **User interface refinement:** Simplifying menus and adding visual cues will make the app more intuitive, especially for first-time users.
- **Security Enhancements:** Stronger encryption and privacy protocols must be prioritized to protect sensitive resident data.
- **Feedback Channels:** A built-in feedback tool will allow residents to share suggestions and report issues directly, ensuring the app grows with community needs.

Future Research Directions

- **Citizen satisfaction studies:** Long-term research can measure how NVgo influences trust and satisfaction with local governance.
- **Comparative analysis:** Comparing NVgo with similar apps in other municipalities will highlight strengths and areas for improvement.
- **Emergency communication:** Exploring NVgo's role in disaster response and crisis management could expand its usefulness.

- **Economic impact:** Studies on cost savings and efficiency gains can help justify scaling NVgo across multiple towns.

Implementation Strategies

- **Training workshops:** Conduct sessions for LGU staff and residents to build confidence in using the app.
- **Pilot testing:** Roll out NVgo in phases, starting with a small group of users before expanding to the entire municipality.
- **Community engagement:** Involve residents in the development process by gathering feedback and making them feel part of the innovation.
- **Monitoring and evaluation:** Establish clear metrics to track NVgo's performance and adjust strategies as needed.

5.5 Impact of the Study

The NVgo proves that innovation is not limited to big cities or national agencies. Even a municipality like Nueva Valencia can embrace digital transformation to solve everyday challenges. This positions the town as a model of forward-thinking

governance, inspiring other municipalities to explore similar solutions.

also NVgo contributes to the growing field of e-governance and mobile solutions. It serves as a case study that future

researchers can build upon, whether in exploring citizen satisfaction, emergency communication.

5.6 Summary

In summary NVgo: Mobile App for Efficient Communication and Service Management in Nueva Valencia was built to address the everyday struggles of residents and local government staff in managing services and communication.

The study showed that NVgo successfully bridges the gap between citizens and their LGU by offering a faster, more transparent, and more reliable way to interact. Residents now have a tool that makes them feel heard, while government staff benefit from reduced paperwork and improved efficiency. The app not only solves technical problems but also strengthens trust and accountability in local governance.